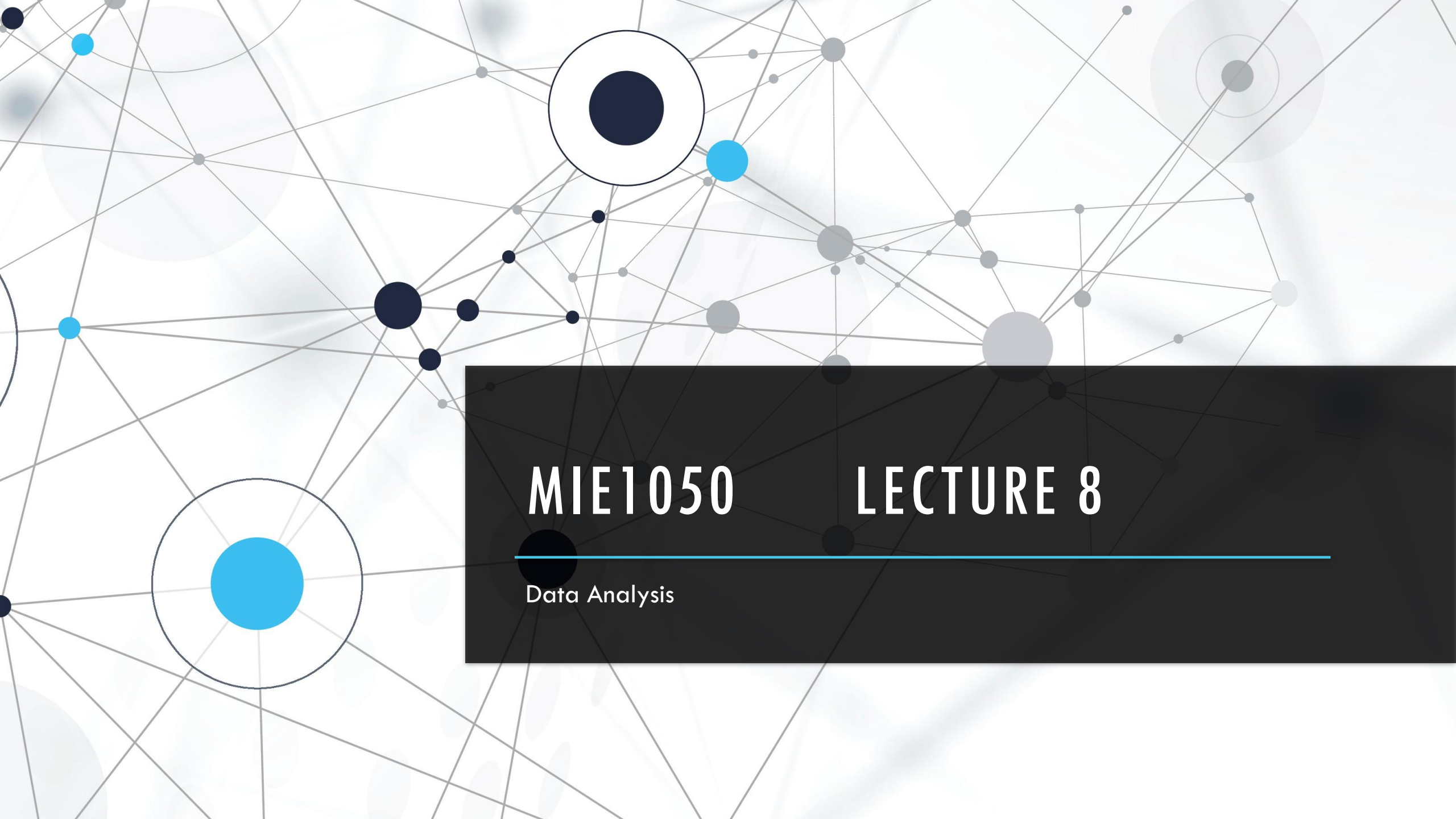


MIE1050 LECTURE 8

Data Analysis

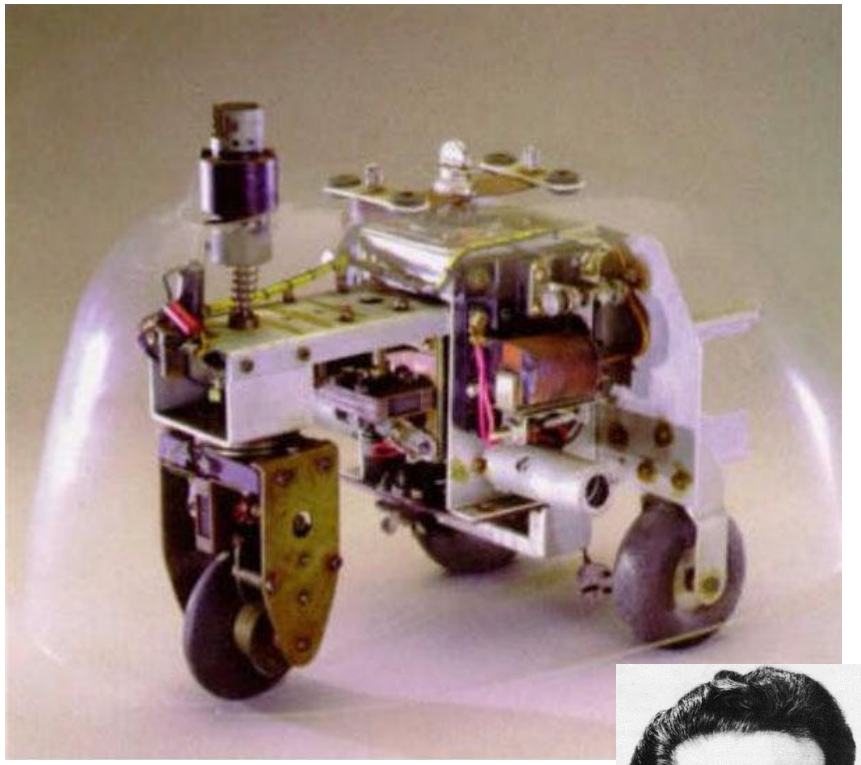
MIE1050H – SCHEDULE

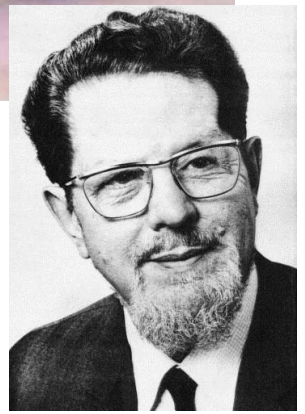
September 9	Lecture	Introduction	
September 16	Lecture	Types of Transducers	
September 23	Lecture	Static vs Dynamic Characteristics	
September 30	Lecture	Linear and non-linear calibration	Group Formation
October 7	Lecture	Signal Acquisition and digitization	
October 14	Lecture	Digital Signal Processing	Range Sensor Calibration Assignment Due
October 21	Lecture	Pattern Analysis, classification, and dimensionality reduction	EEG Design Assignment Due/Project Proposal Due
October 28	No Lecture	Reading Week (no classes)	
November 4	Lecture	Sensor Fusion – Spatial and Temporal Data fusion	Real-time Digital Filtering Assignment Due
November 11	Lecture	Continuous Sensor Data Analytics	Classsification Assignment Due
November 18	Lecture	Wireless sensors and IOT devices	Multi-Sensor Sensor Network Assignment
November 25	Lecture / Tutorial	Intelligent Sensor Networks	
December 2	Project	Project Presentations	
December 9	Project	Project Presentations	
December 16	No Lecture		Final Project Report Due

ORIGIN OF ARTIFICIAL INTELLIGENCE

First Artificial Intelligent Circuit

- Developing a practical EEG instrument (invented originally by Hans Berger), Mr. H Grey Walter discovered Alpha brain waves.
- In 1948 Dr. Walter built a mechanical turtle that mimicked animal like behaviour using a neural-like circuit.

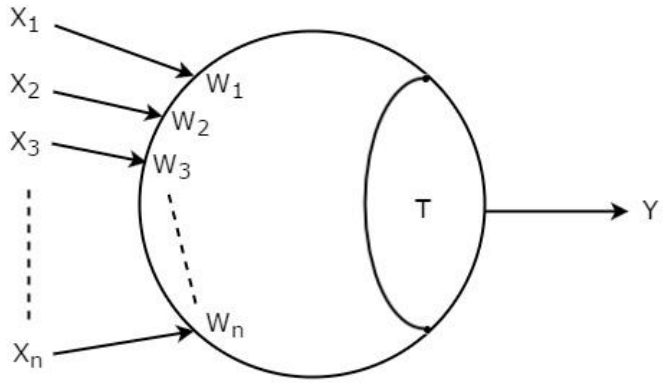




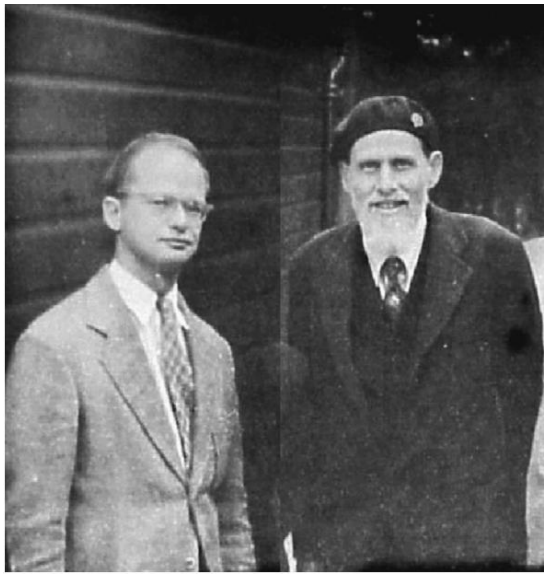
ORIGIN OF ARTIFICIAL INTELLIGENCE

First Mathematical Neural Network:

- In 1943, Warren McCulloch and Walter Pitts published "A Logical Calculus of the ideas Immanent in Nervous Activity," describing the first mathematical model of a neural network.
- Building on ideas in Alan Turing's "On computable Numbers", their paper showed that simple elements connected in a neural network can have immense computational power.



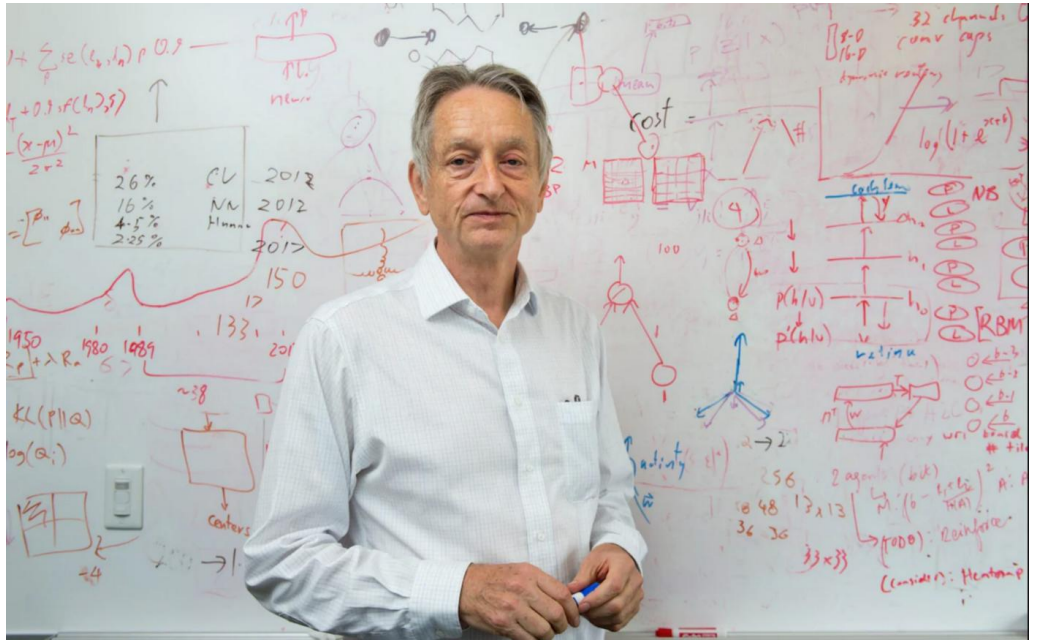
Threshold Gate



ORIGIN OF ARTIFICIAL INTELLIGENCE

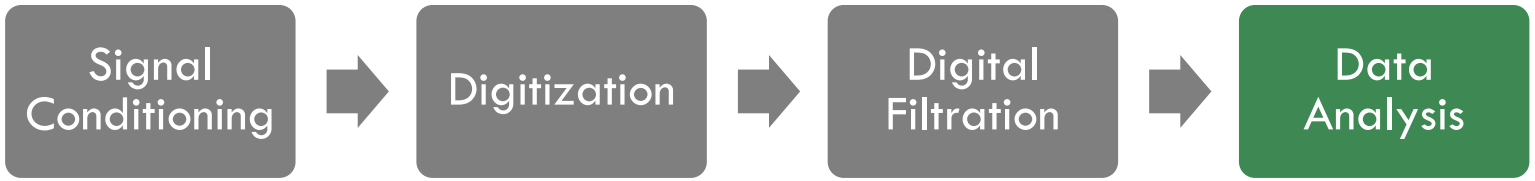
Deep Learning pioneer:

- Professor Geoffrey Hinton, University of Toronto, was one of the pioneers of applying back propagation to a multi-layer neural network.
- Developed in 1980s and more widely accepted in early 2000s, deep learning learns higher-level features from the raw input.
- Professor Hinton won the Nobel prize in physics in 2024 for his contribution in deep learning in AI.



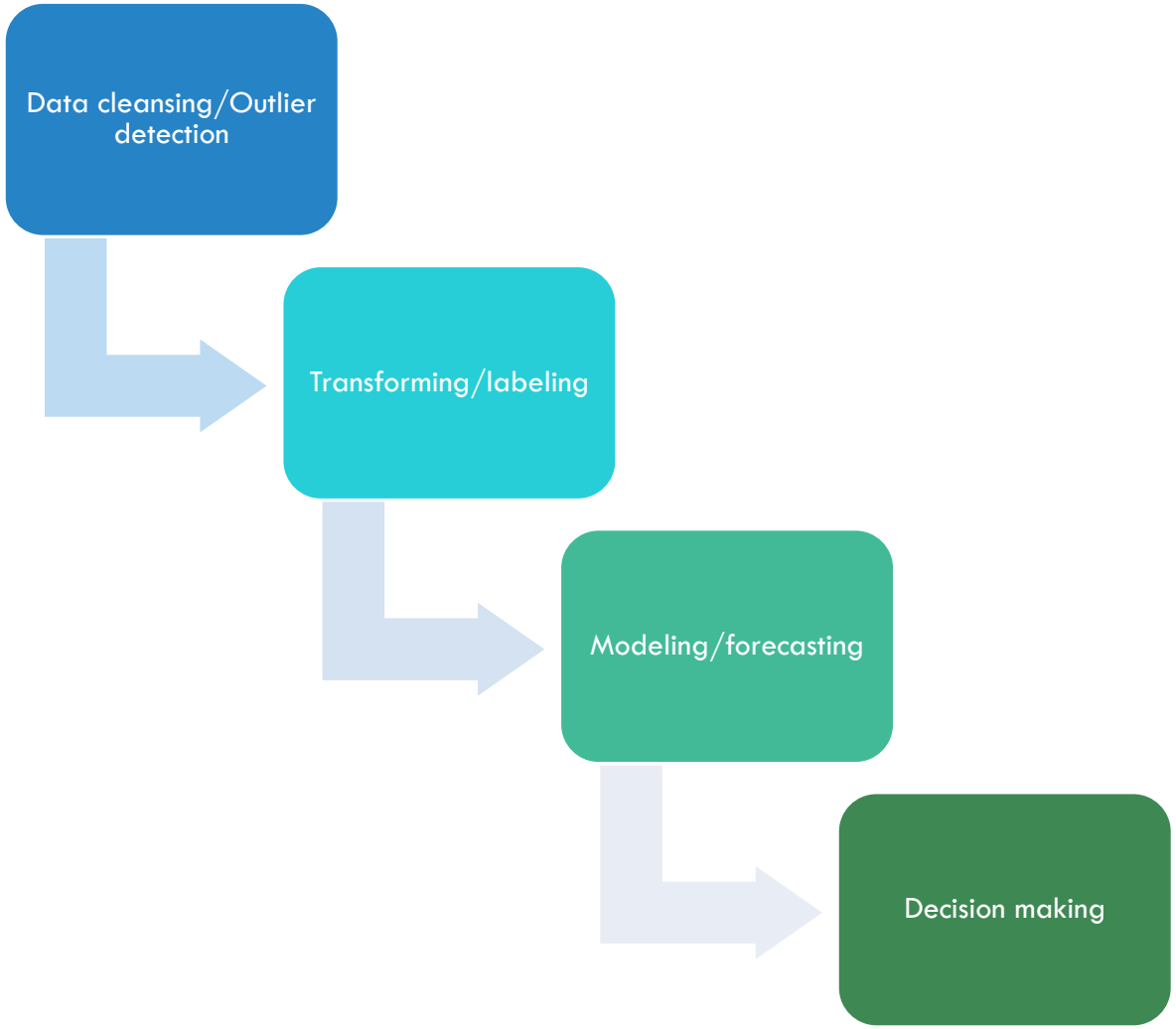
SIGNAL PROCESSING

- Once a signal from a sensor has been conditioned, digitized and filtered, it can be analyzed.



DATA ANALYSIS

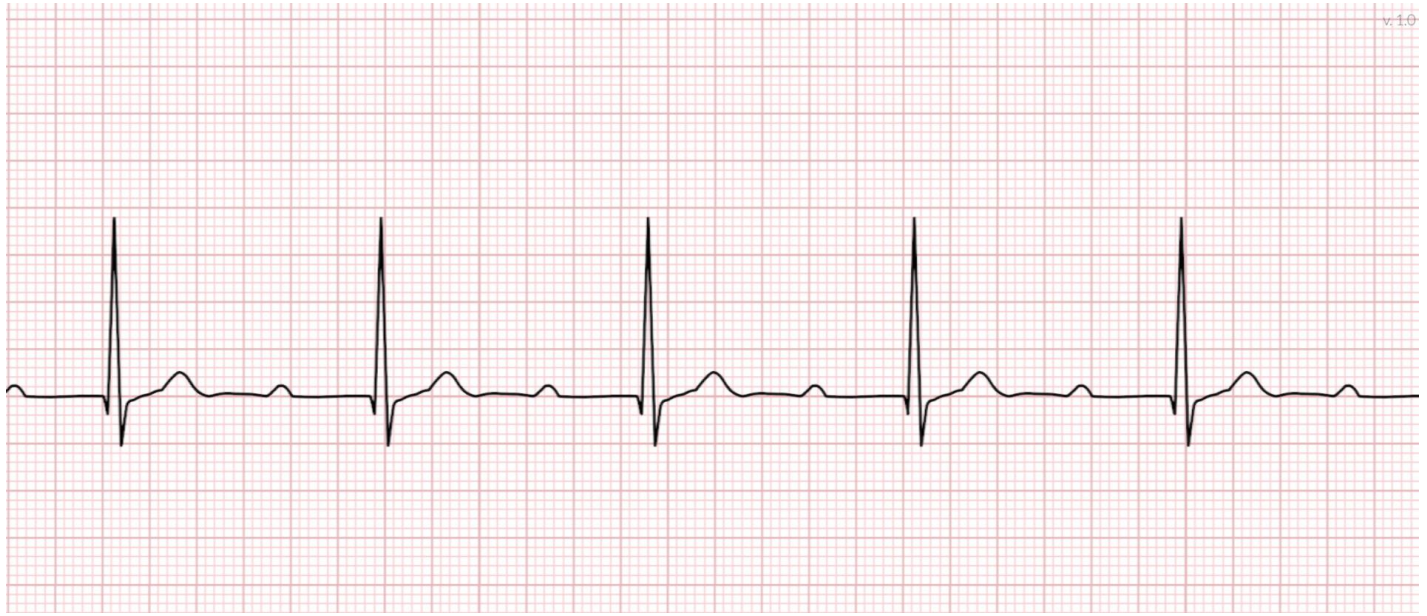
- Data analysis refers to the broad field of post filtration data processing :
- In this course we will focus on pattern recognition for transforming/labeling the data.

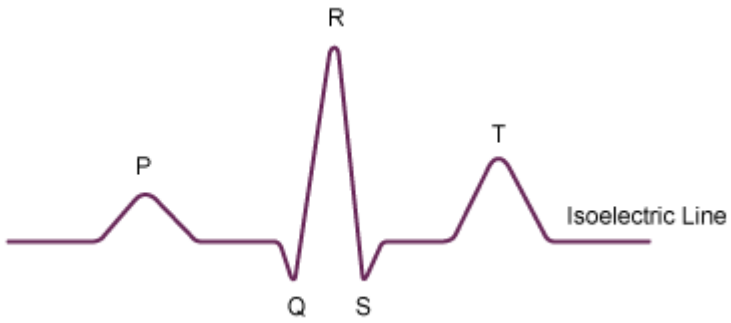


PATTERN RECOGNITION

Pattern recognition is

- the discovery of regularities in the data
- the use of these regularities to classify data into categories.





AI VS ML VS PR

What is the difference between:

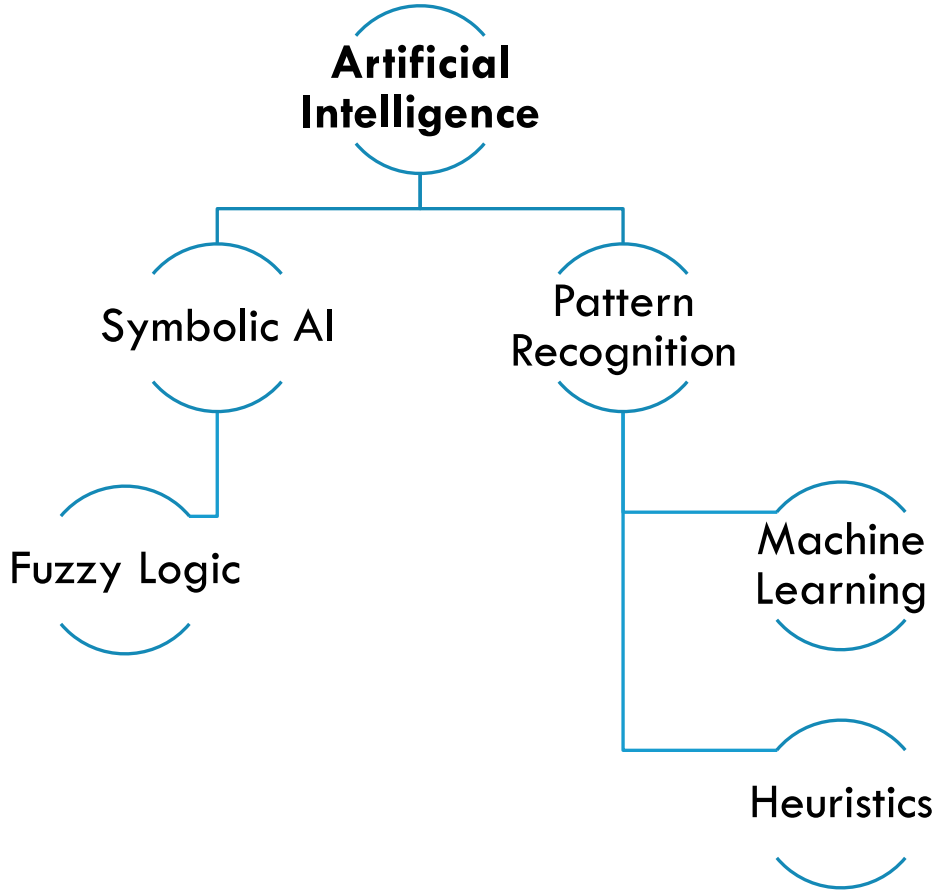
1. Machine Learning,
2. Artificial Intelligence, and
3. Pattern Recognition?

ARTIFICIAL INTELLIGENCE

Artificial intelligence (AI) refers to systems that mimic humans in learning and problem solving.

The aim is to make intelligent **decisions** based on acquired and applied knowledge

AI can be realized through Pattern recognition or Symbolic AI.

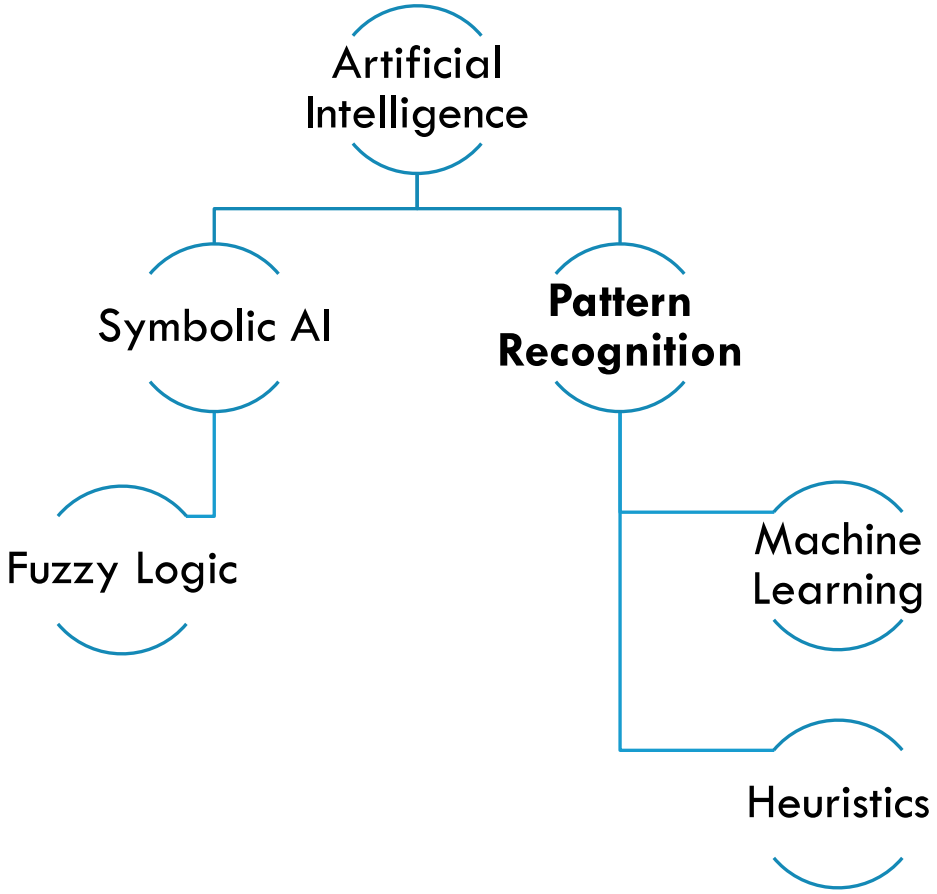


PATTERN RECOGNITION

Pattern Recognition refers to the automated recognition of regularities in data.

The aim is to take action based on these regularities such as classification of the data into different categories.

Machine Learning and Heuristics (hand-crafted rules) are a form of pattern recognition.

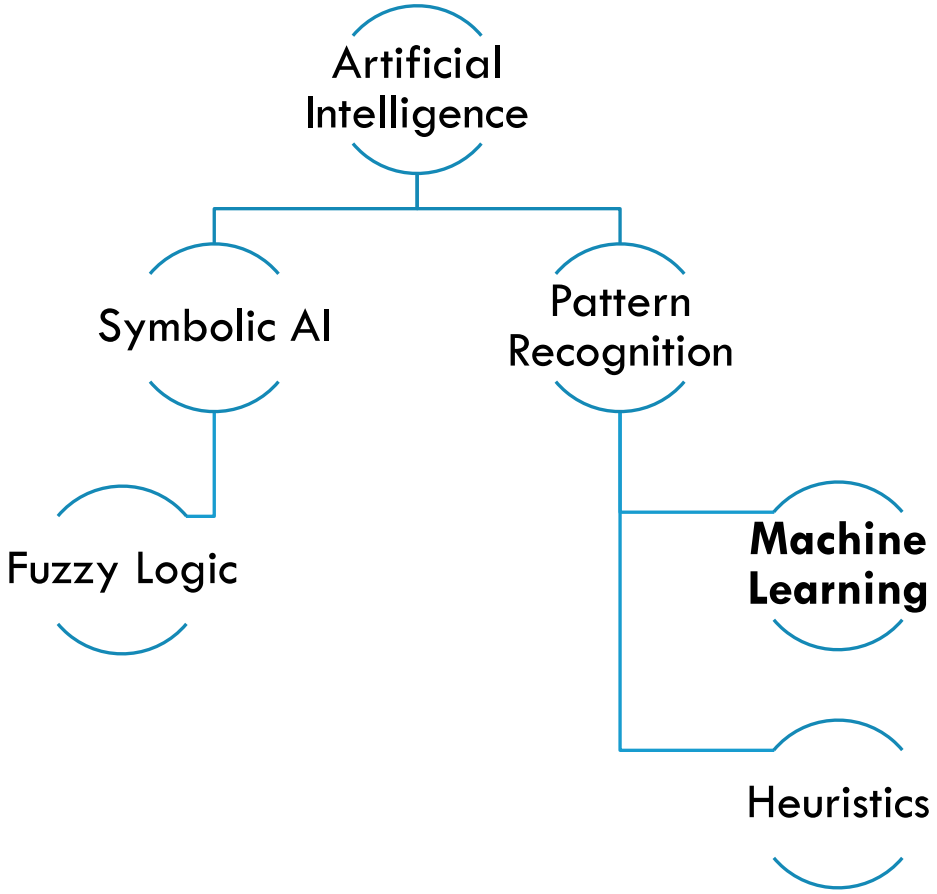


MACHINE LEARNING

Machine Learning (ML) refers to the process of a system or program that acquired data and learns from it without being explicitly program.

The goal is to learn from data on certain task to maximize the performance of machine on this task.

ML is a form of pattern recognition.

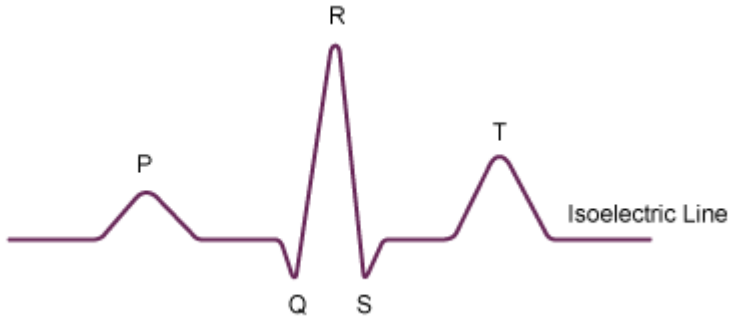


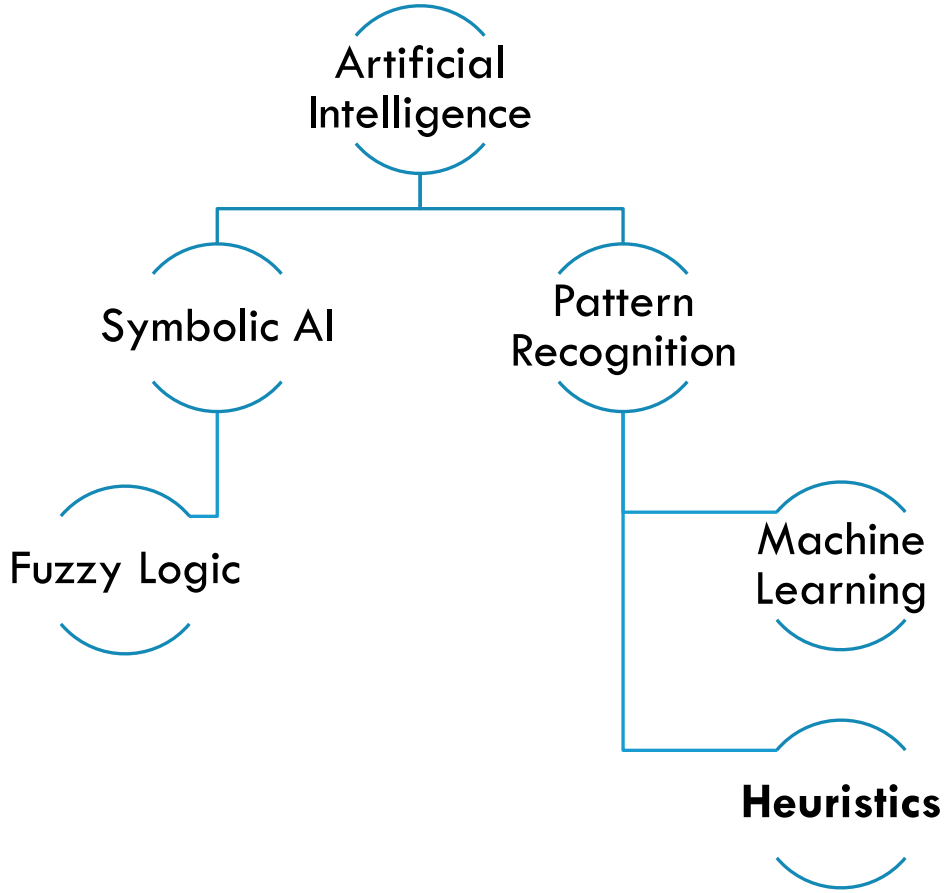
HEURISTICS

Heuristics refers to type of pattern recognition when the designer inputs a set of predefined rules when categorizing data.

For example

- any value greater than x will be classified as R.
- If R is detected then any value greater than X2 is classified as T.





SUPERVISED VS UNSUPERVISED LEARNING

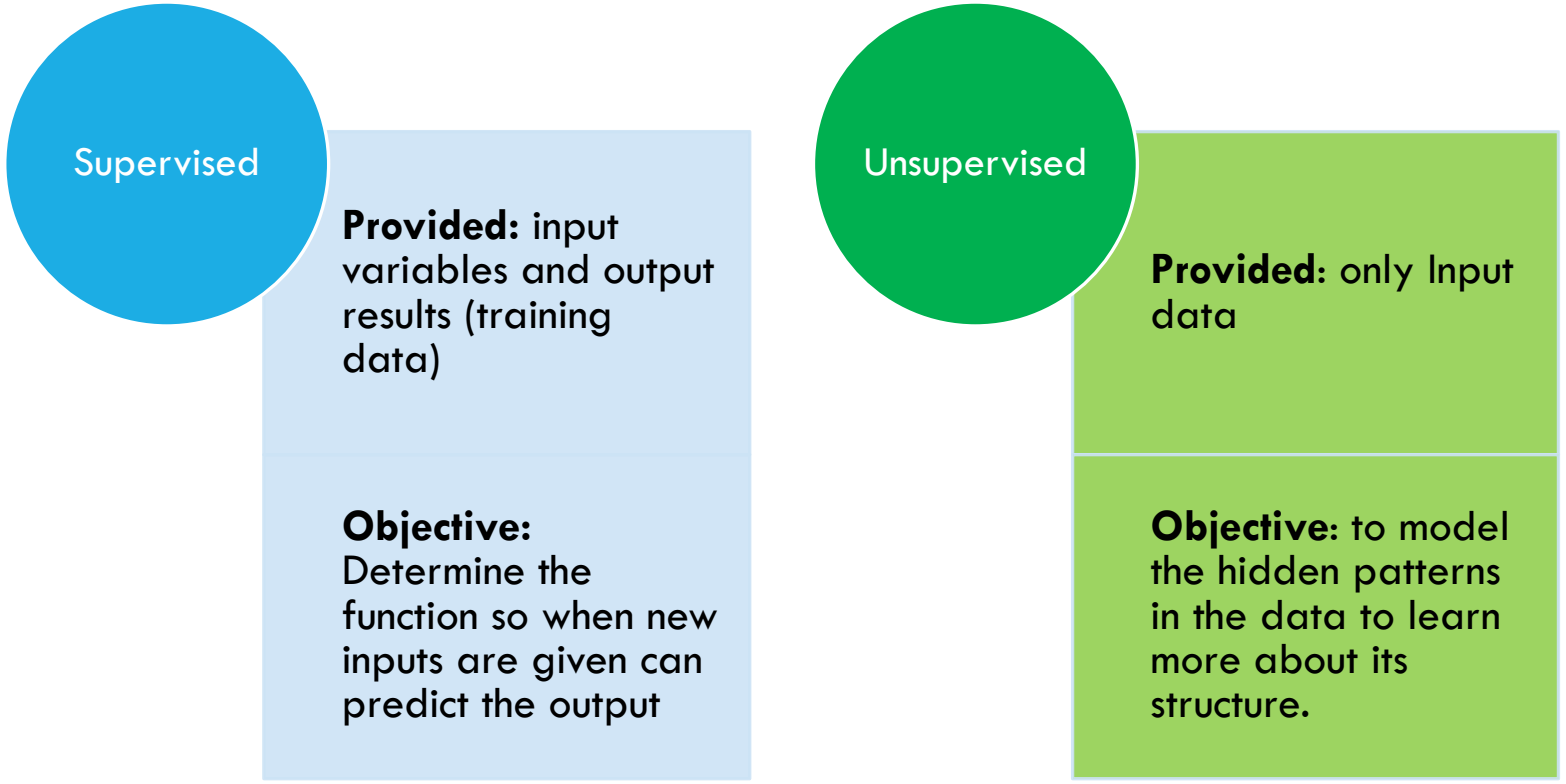
For this course we will focus **only** on machine learning as a form of pattern recognition.

The objective is to provide methodology for implementing ML for

- Detection of outliers,
- Dimensionality reduction,
- Classification and categorization.

SUPERVISED VS UNSUPERVISED LEARNING

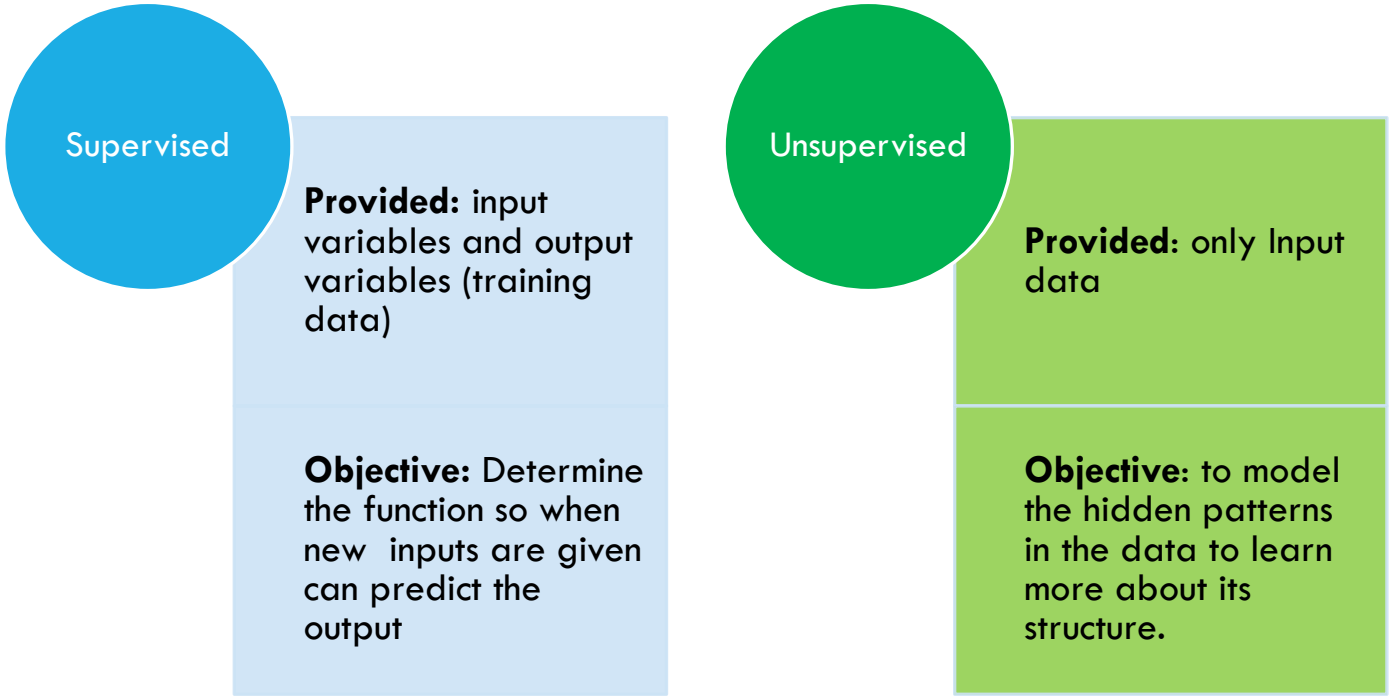
Pattern recognition or ML algorithms can be categorized as either Supervised or Unsupervised learning.



SUPERVISED VS UNSUPERVISED LEARNING

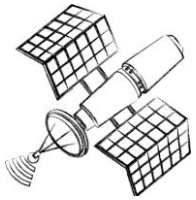
Questions:

- Would linear regression (line fitting) be considered a form of ML?
- Would Kalman Filter be considered ML?



KALMAN FILTER VS MACHINE LEARNING

Aspect	Kalman Filter	Machine Learning
Approach	Model-based (requires explicit mathematical model of the system)	Data-driven (learns patterns from data)
Adaptability	Parameters usually fixed or manually tuned	Learns parameters automatically from training data
Objective	Estimate the hidden state of a known model	Learn unknown relationships between inputs and outputs
Examples	GPS tracking, sensor fusion, robotics navigation	Image classification, regression, natural language processing



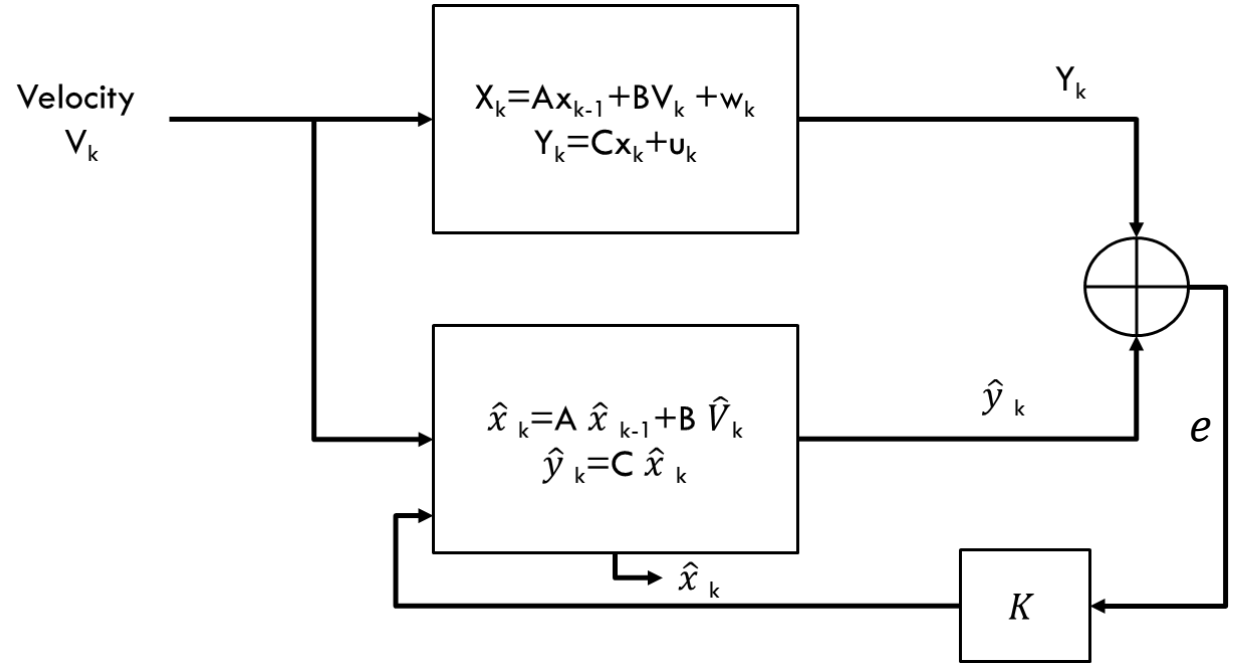
KALMAN FILTER FOR OUR PRACTICAL EXAMPLE

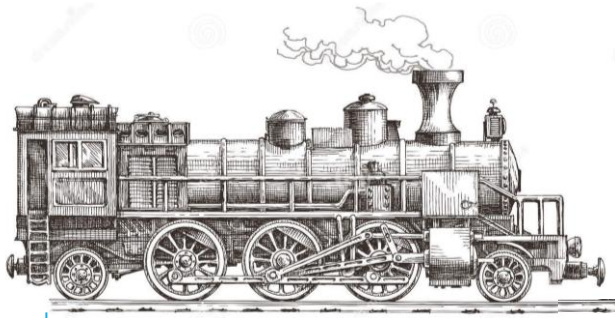
Kalman Filter has 2 steps: predict and update.

$$\hat{x}_k = A\,\hat{x}_{k-1} + B\,\hat{V}_k + K_k[Y_k - C(A\,\hat{x}_{k-1} + B\,\hat{V}_k)]$$

$A\,\hat{x}_{k-1} + B\,\hat{V}_k$: *Predict*

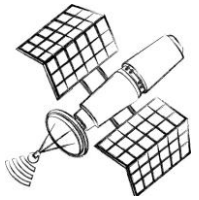
$K_k[Y_k - C(A\,\hat{x}_{k-1} + B\,\hat{V}_k)]$: Update





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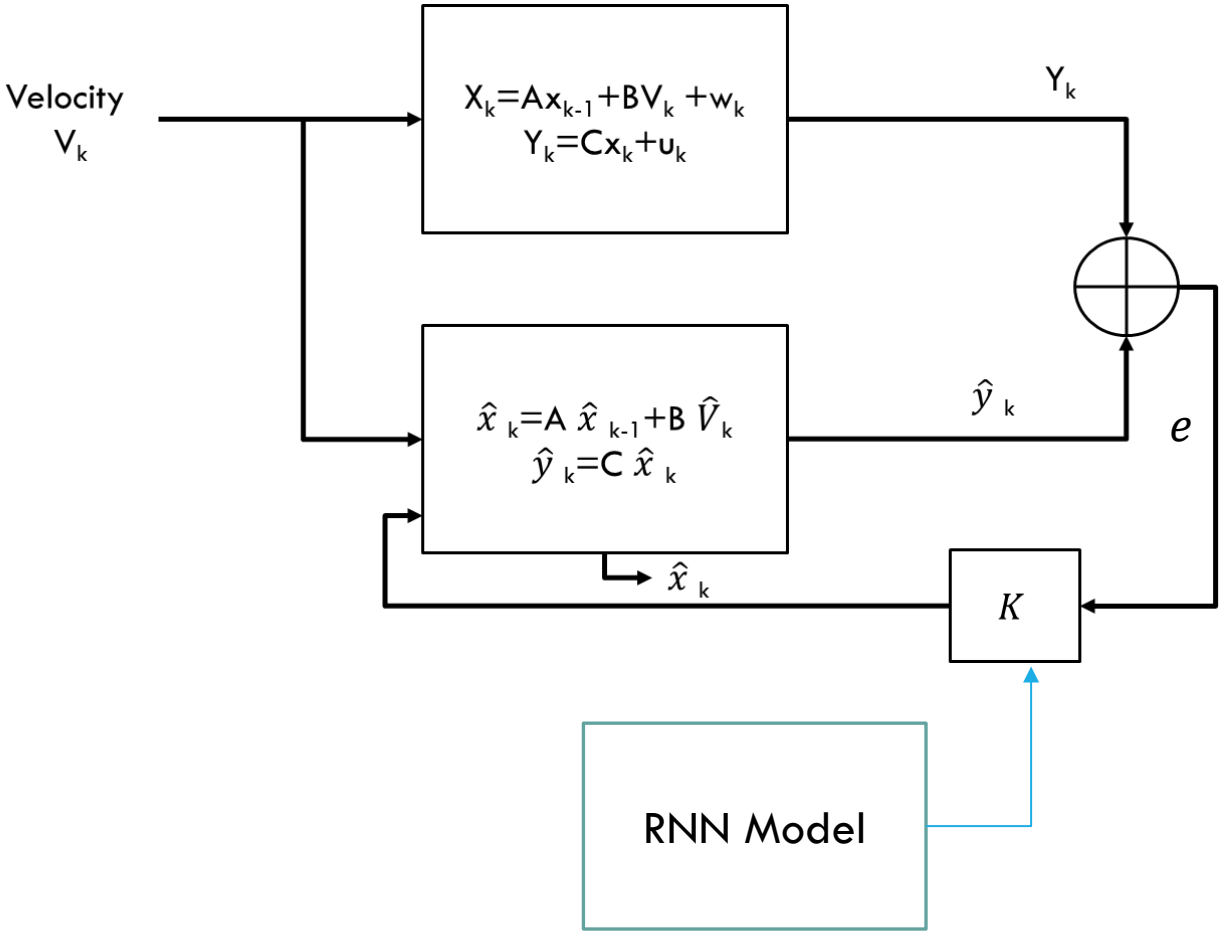


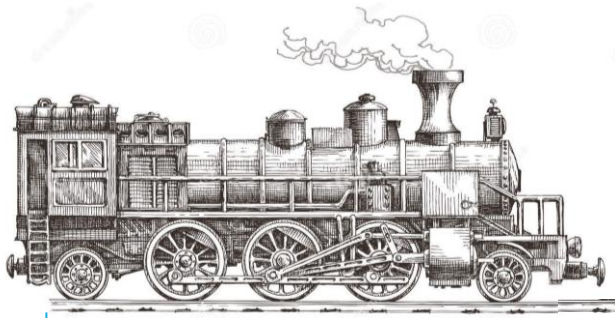
KALMAN-NET FILTER

Kalman-Net has 3 steps: predict, Estimate parameters and update.

$$\hat{x}_k = A\,\hat{x}_{k-1} + B\,\hat{V}_k + K_k[Y_k - C(A\,\hat{x}_{k-1} + B\,\hat{V}_k)]$$

Predict Update

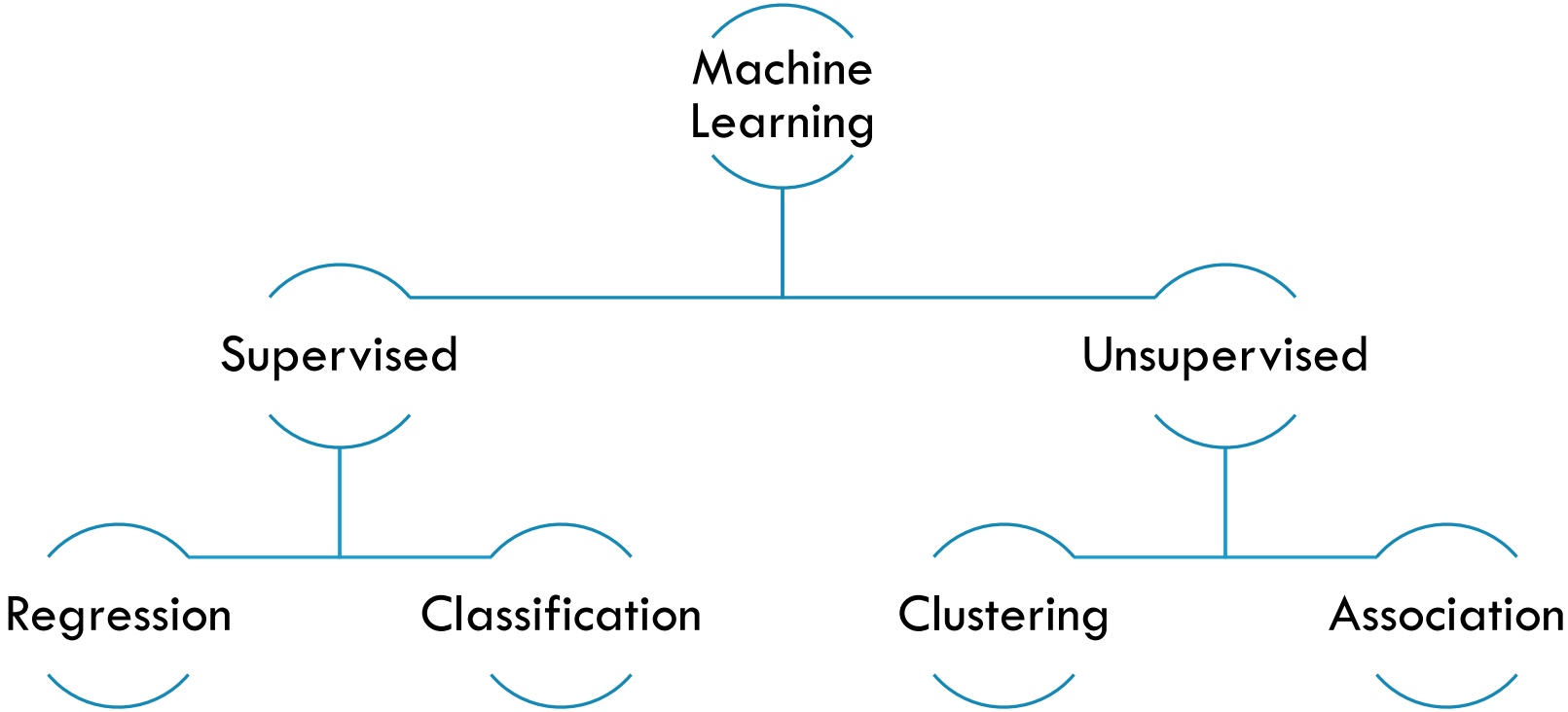




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SUPERVISED VS UNSUPERVISED LEARNING

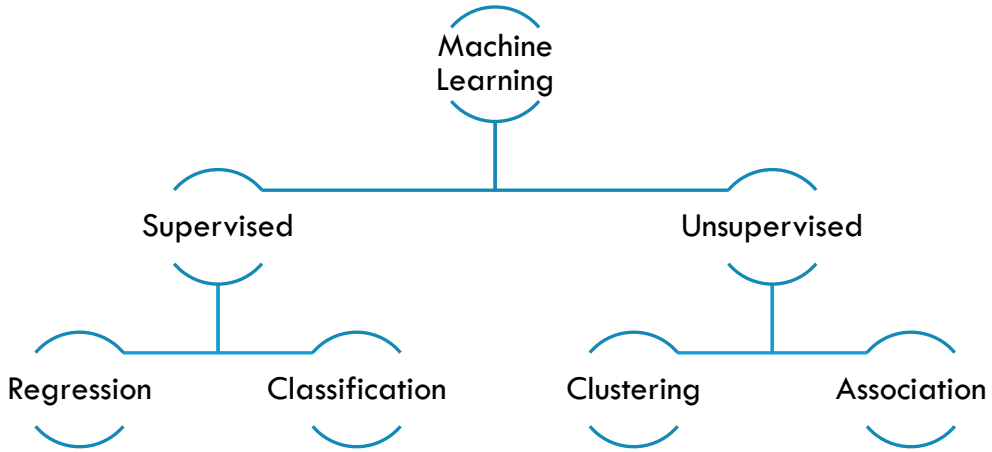


SUPERVISED LEARNING

Supervised Learning can be categorized as either:

Regression: model to have continuous input and continuous output.

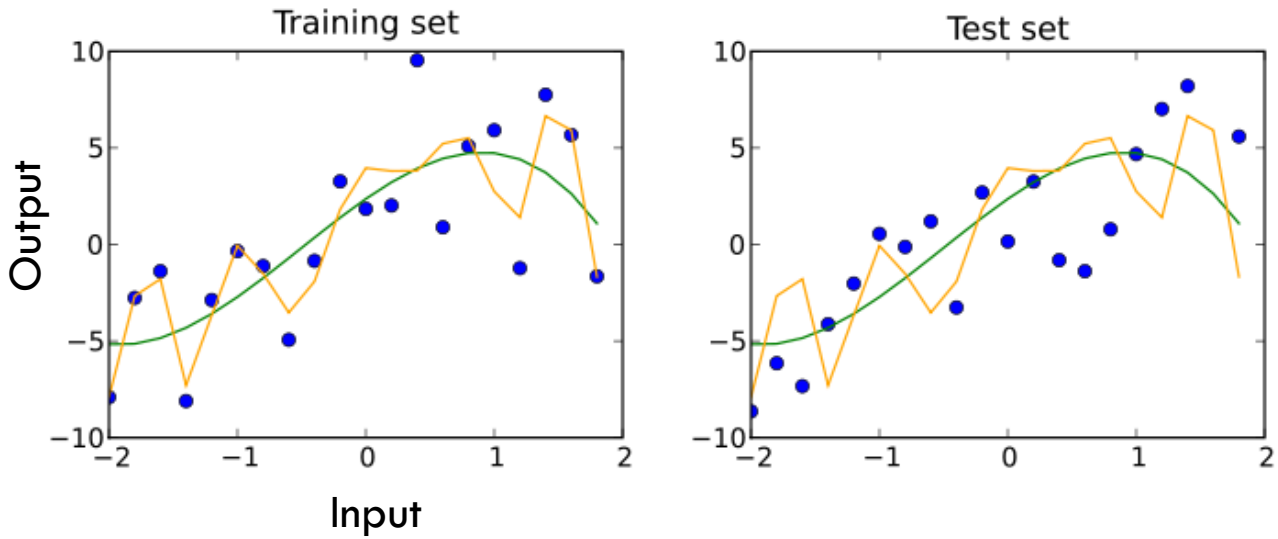
Classification: continuous or discrete input but ONLY discrete output (categories)



SUPERVISED LEARNING

In supervised machine learning there are 3 set of data that are used:

1. Training data set
   - Should have both input and output
   - For classification it should be pre-labeled
2. Verification data set
3. Test data set



	No.	Outlook	Temperature	Humidity	Play Golf
Training	1	sunny	hot	normal	N
	2	rain	hot	high	N
	3	overcast	hot	high	Y
	4	rain	mild	high	Y
	5	rain	cool	high	Y
Verification	6	sunny	mild	normal	N
	7	overcast	cool	normal	Y
Test	8	sunny	mild	high	?

FEATURE SPACE

- The input into any learning algorithm is called a feature.
- Feature can be
  - any distinctive aspect, quality, or characteristics
  - Symbolic (e.g. colour) or Numerical (e.g. length)

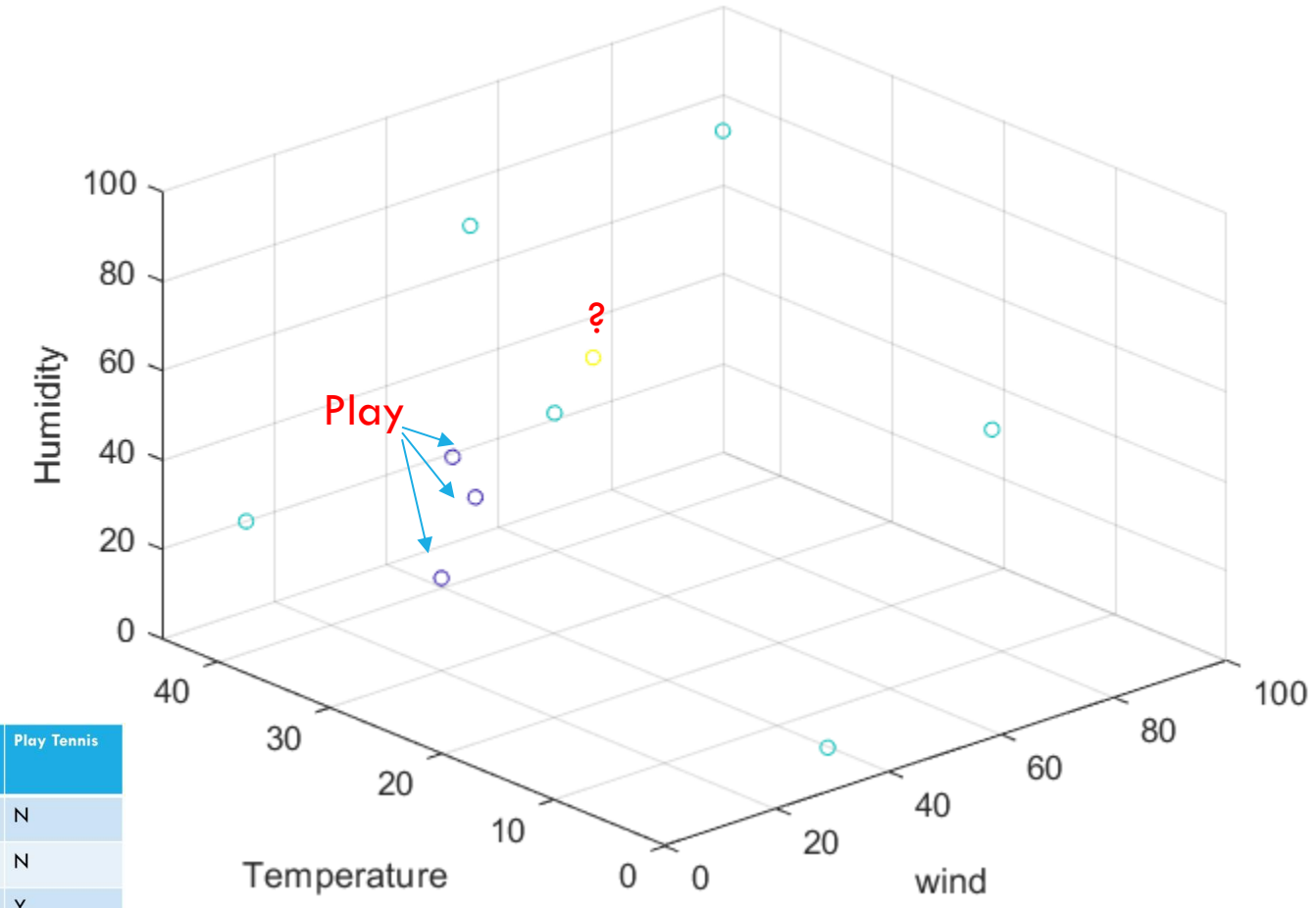
What are the Features in this example?

No.	Wind (Km/hr)	Temperature ˚C	Humidity %	Play Tennis
1	50	35	40	N
2	70	30	100	N
3	10	25	30	Y
4	15	45	20	N
5	100	21	30	N
6	22	28	40	Y
7	33	34	90	N
8	8	23	60	Y
9	29	0	10	N
10	**35**	**24**	**70**	**?**

FEATURE SPACE

- The combination of the d features can be considered as a d-dimensional vector called "Feature Vector".
- The feature vectors can be represented as points in a d-dimensional space called "Feature Space"

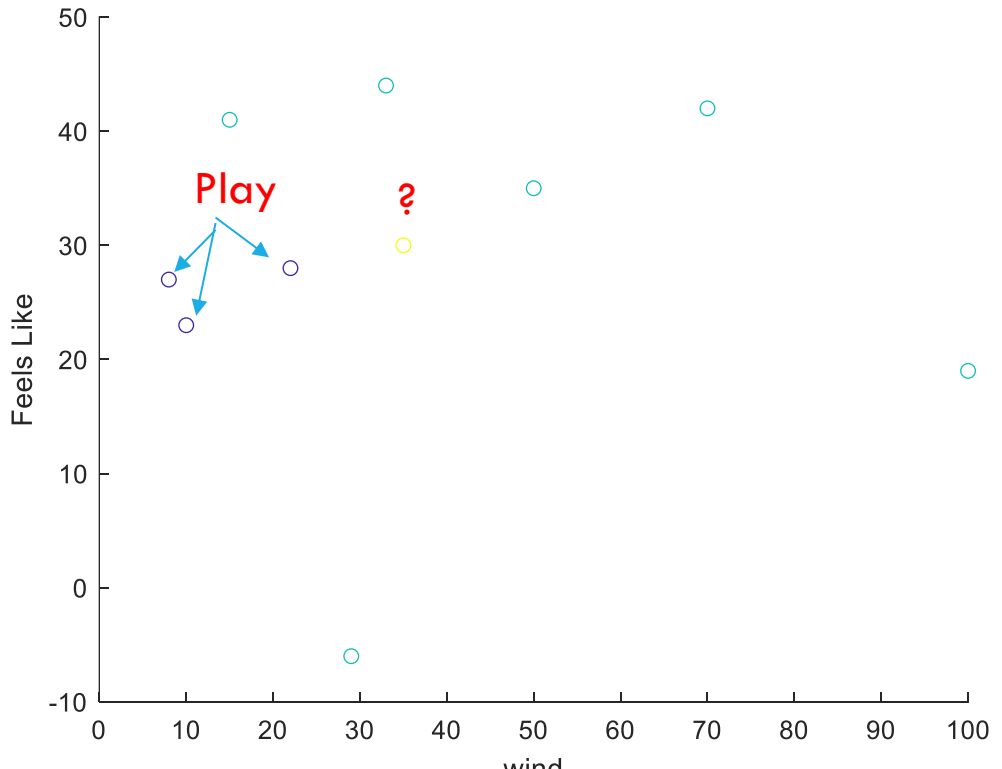
No.	Wind (Km/hr)	Temperature °C	Humidity %	Play Tennis
1	50	35	40	N
2	70	30	100	N
3	10	25	30	Y
4	15	45	20	N
5	100	21	30	N
6	22	28	40	Y
7	33	34	90	N
8	8	23	60	Y
9	29	0	10	N
10	**35**	**24**	**70**	**?**



FEATURE SPACE

- To better visualize and reduce computational complexity, we can reduce the number of features by combining T and RH to a “feels like” metric.

No.	Wind (Km/hr)	Feels like	Play Tennis
1	50	35	N
2	70	42	N
3	10	23	Y
4	15	41	N
5	100	19	N
6	22	28	Y
7	33	44	N
8	8	27	Y
9	29	-6	N
10	**35**	30	**?**

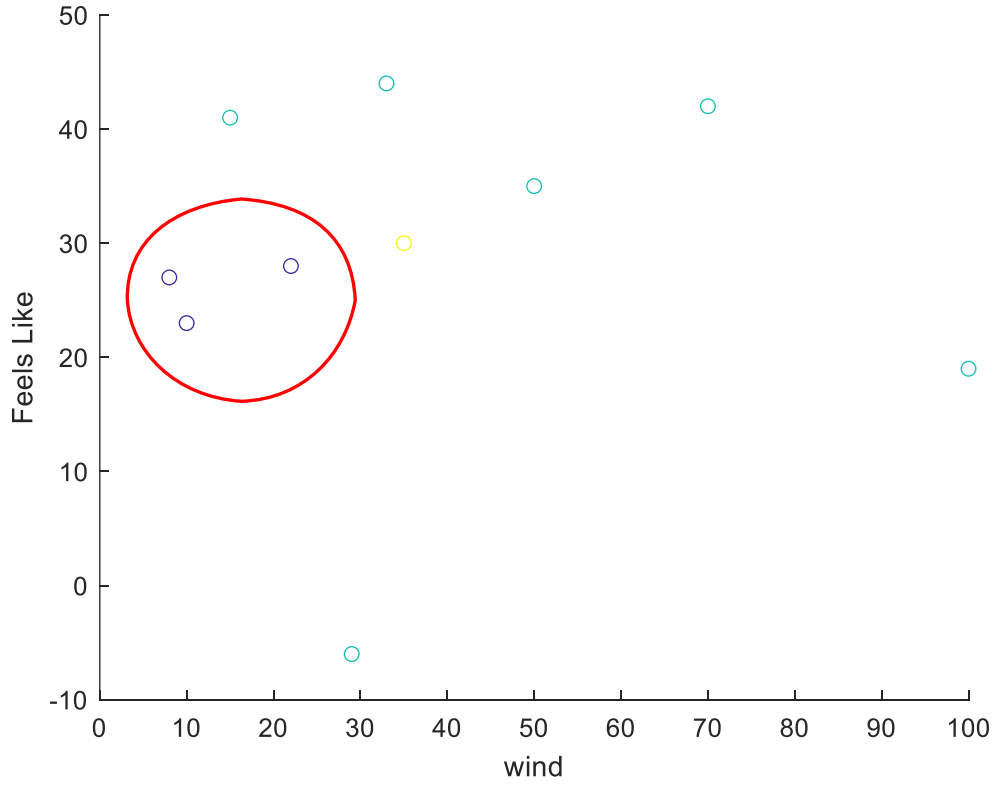


K-NEAREST NEIGHBOR

K-Nearest Neighbor (KNN) is a form of supervised classification and regression algorithm.

The Assumption:

Similar data points are close to each other in feature space.



K-NEAREST NEIGHBOR

For each point of the data-set a distance is calculated.

Distance can be a simple Euclidian distance:

$$D_i = \mathrm{Sqrt}[(X_i - X')^2 + (Y_i - Y')^2]$$

Where X_i and Y_i are the feature values of point i of the data-set and X' and Y' are the feature values of the test point.

No.	Wind (Km/hr)	Feels like	Play Tennis	Distance
1	50	35	N	16
2	70	42	N	37
3	10	23	Y	26
4	15	41	N	23
5	100	19	N	66
6	22	28	Y	13
7	33	44	N	14
8	8	27	Y	27
9	29	-6	N	36
10	**35**	30	**?**	0

K-NEAREST NEIGHBOR

- For classification the mode of the K nearest data set is used.
  - For K=1 or nearest neighbor estimation the answer will be YES play (point 6).
  - The nearest neighbor (K=1) can be subjected to error based on a single data point.
  - For K=3 the answer will be NO as there are more NO answers than Yes (2 vs 1)
  - Usually K is selected to be an odd number.

No.	Wind (Km/hr)	Feels like	Play Tennis	Distance
6	22	28	Y	13
7	33	44	N	14
1	50	35	N	16
4	15	41	N	23
3	10	23	Y	26
8	8	27	Y	27
9	29	-6	N	36
2	70	42	N	37
5	100	19	N	66

K-NEAREST NEIGHBOR

The issue with our current example is that we calculate distance without considering the range of the data in the feature.

If X feature would have a range of 0 to 10 and Y from -1000 to + 1000, Y feature would dominate the distance calculation.

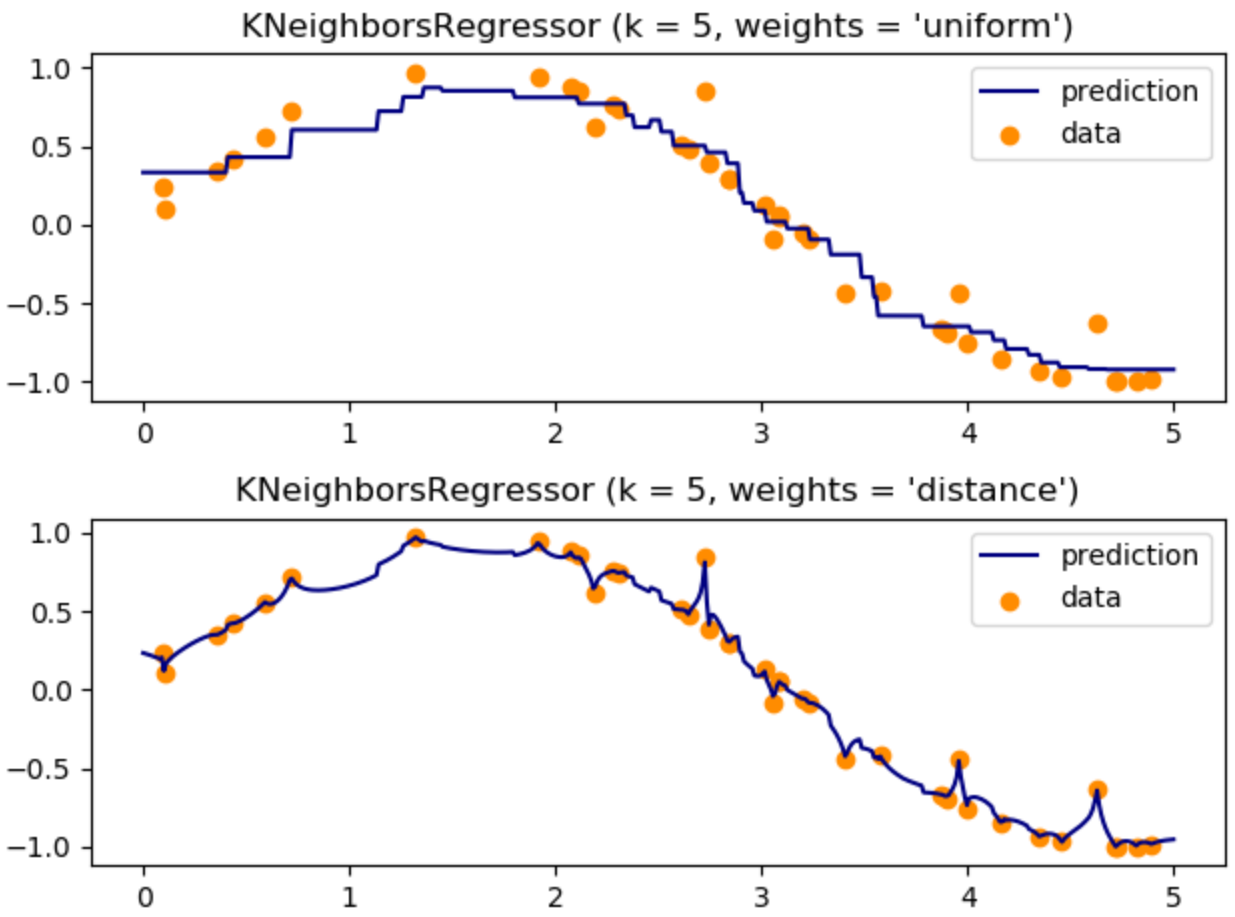
To remove this effect all data should be scaled to be similar range (e.g. 0 and 1).

No.	Wind (Km/hr)	Feels like	Play Tennis	Distance
6	0.22	0.46	Y	0.134
1	0.5	0.58	N	0.172
7	0.33	0.733	N	0.234
4	0.15	0.683	N	0.271
8	0.08	0.45	Y	0.275
3	0.1	0.383	Y	0.276
2	0.7	0.7	N	0.403
9	0.29	-0.1	N	0.603
5	1	0.316	N	0.675

K-NEAREST NEIGHBOR

- To implement KNN for regression analysis:
- The same algorithm is used but instead of returning the “mode”, it returns the “mean” of the values of the closest K data sets.
- To provide better prediction a weighted average is used where the weights are based on the distance calculated.

$$Y' = \frac{\sum Y_i d_i}{\sum d_i}$$



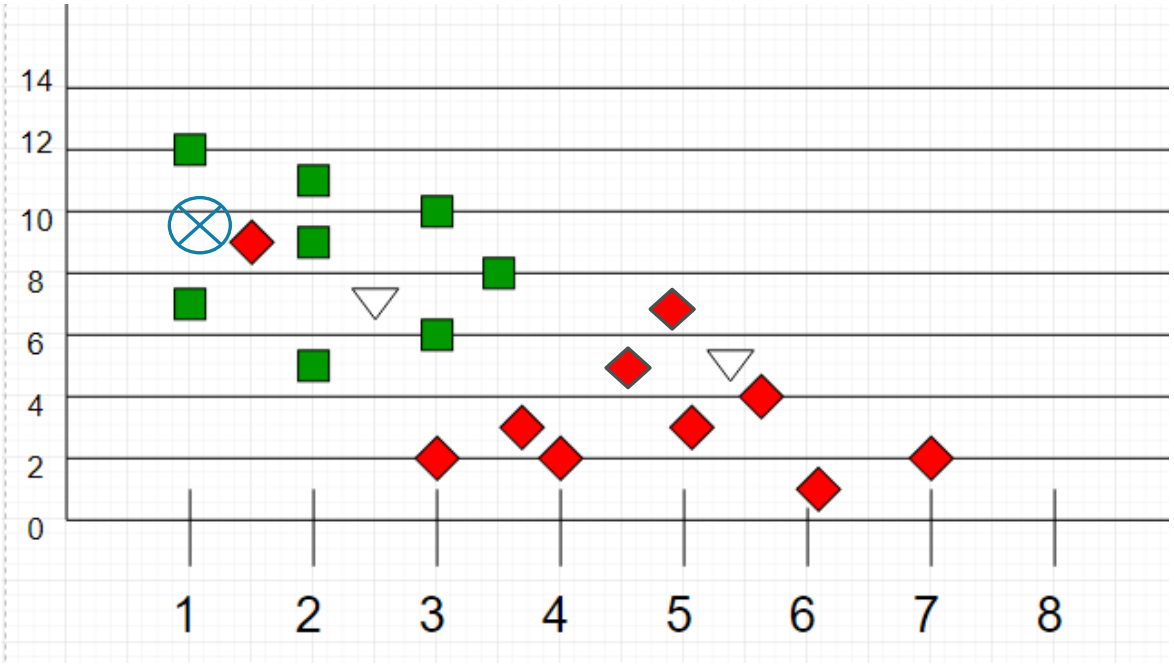
K-NEAREST NEIGHBOR

Choosing the right value of K

Decreasing the value of K to 1 the results will be more unstable (dependent on a single point of data).

Increasing the value makes the system more stable and less sensitive to single data point.

Increasing the K value too greatly will make the system less responsive (i.e. all data points get the same classification).



For K=1 →
For K=3 →
For K=17→

K-NEAREST NEIGHBOR

Advantage:

1. The algorithm is simple and easy to implement.
2. There's no need to build a model, tune several parameters, or make additional assumptions.
3. The algorithm is versatile. It can be used for classification and regression.

Disadvantage:

1. The algorithm gets significantly slower as the number of samples and/or features increase

BAYES THEOREM

- Naïve Bayes Classifier (NBC) provides a more computationally efficient and more accurate classification than KNN.
- To be able to implement an NBC we first need to understand Bayes Theorem.
- Developed in 1740s by Reverend Thomas Bayes and later rediscovered independently by Pierre Simon Laplace.
- The theorem provides the conditional probability of an event given prior knowledge of other related circumstances.



BAYES THEOREM

Example

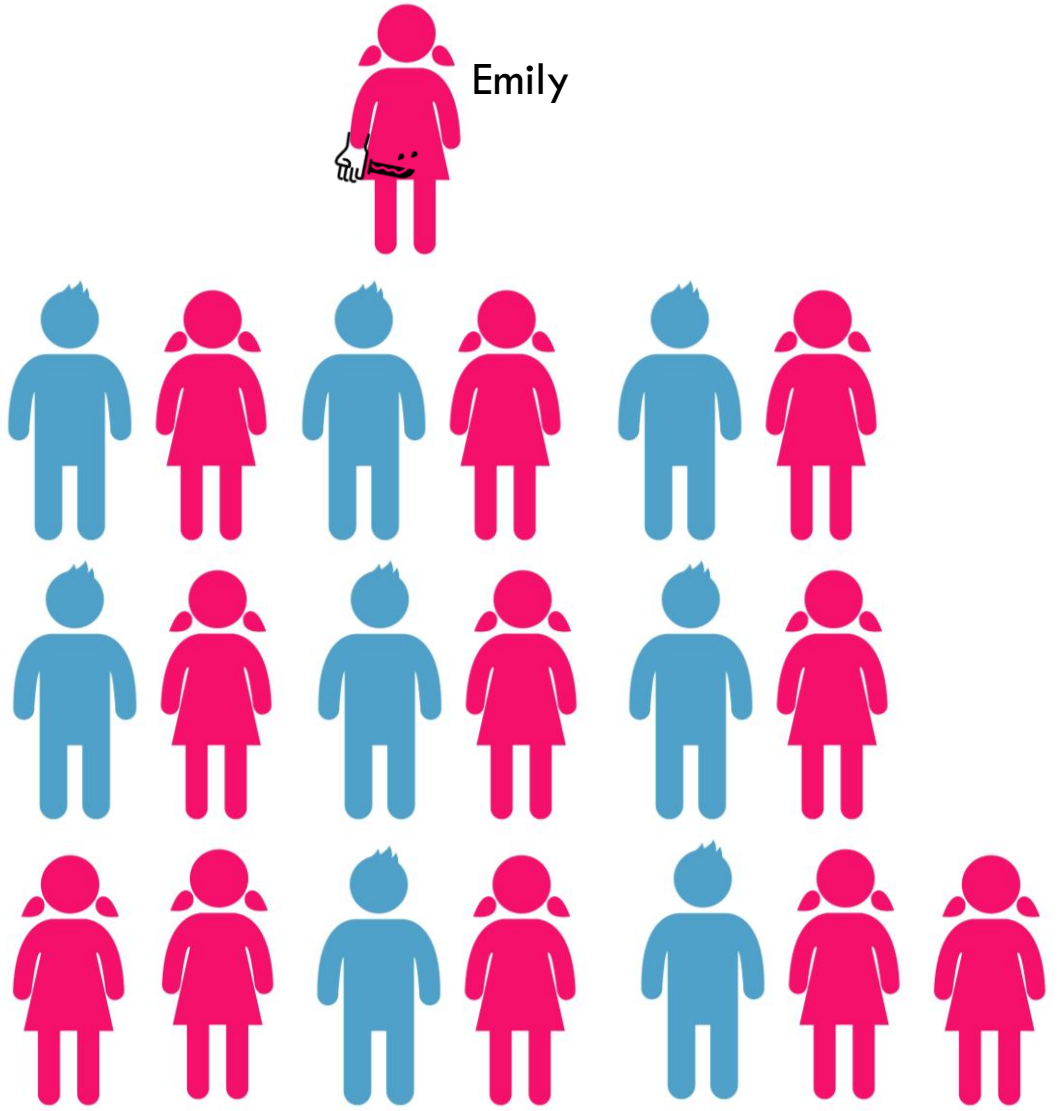
- You are participating in a murder mystery party.
- There are 20 guests (not including yourself).
- 12 girls and 8 boys.



BAYES THEOREM

Example

- What is the probability of Emily being the murderer?
  - P(Emily_is_Murderer)= 1/20 = 0.05
  - This is called the prior probability as it is based on no other conditions.
- If you discover that the murderer is a girl what is the new probability?
  - P(Emily_is_Murderer|Murderer_is_girl)=1/12=0.083
- The "|" represents conditional probability
- What is the probability of Emily being the murderer if the murderer is a boy?



BAYES THEOREM

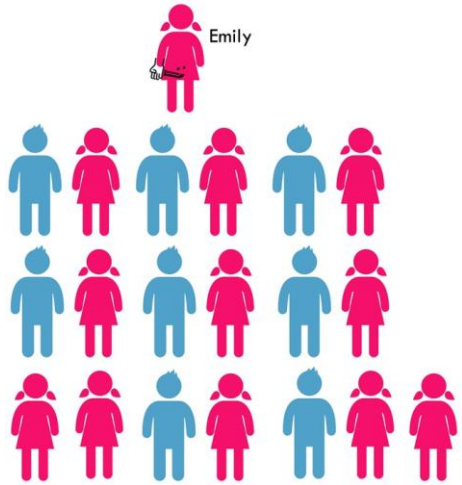
- Sometimes we don't know how a new piece of evidence will alter our conditional probability.
- This is what Bayes theorem is used to calculate.

$$P(H|E) = P(E|H)*P(H)/P(E)$$

Where

- P(H) is prior probability of the Hypothesis
- P(E) is prior probability of related evidence

BAYES THEOREM



- New Evidence is found:
  - Emily has a weapon!
- What is the Probability of Emily being the Murderer and the murderer is a girl?

$P(H|E) = P(E|H)*P(H)/P(E)$

$$P(\text{Emily_is_Murderer}|\text{She_has_weapon})=\frac{P(\text{She_has_weapon}|\text{She_is_murderer})*P(\text{Emily_is_Murderer}|\text{Murderer_is_girl})}{P(\text{Emily_has_weapon})}$$

  - Assume there are 5 weapons in the game.

$P(\text{Emily_has_weapon})=5/20=0.25$

And assuming that murderer needs a weapon

$P(\text{She_has_weapon}|\text{She_is_murderer})=1$

$P(\text{Emily_is_Murderer}|\text{She_has_weapon})=1*0.083/0.25=0.33$ or 33%

NAÏVE BAYES CLASSIFIER

We wish to create a system that would detect which elevator we are riding in based on:

1. Acceleration
2. Deceleration

Note: the acceleration and deceleration of the elevator is affected by the number of people in the elevator

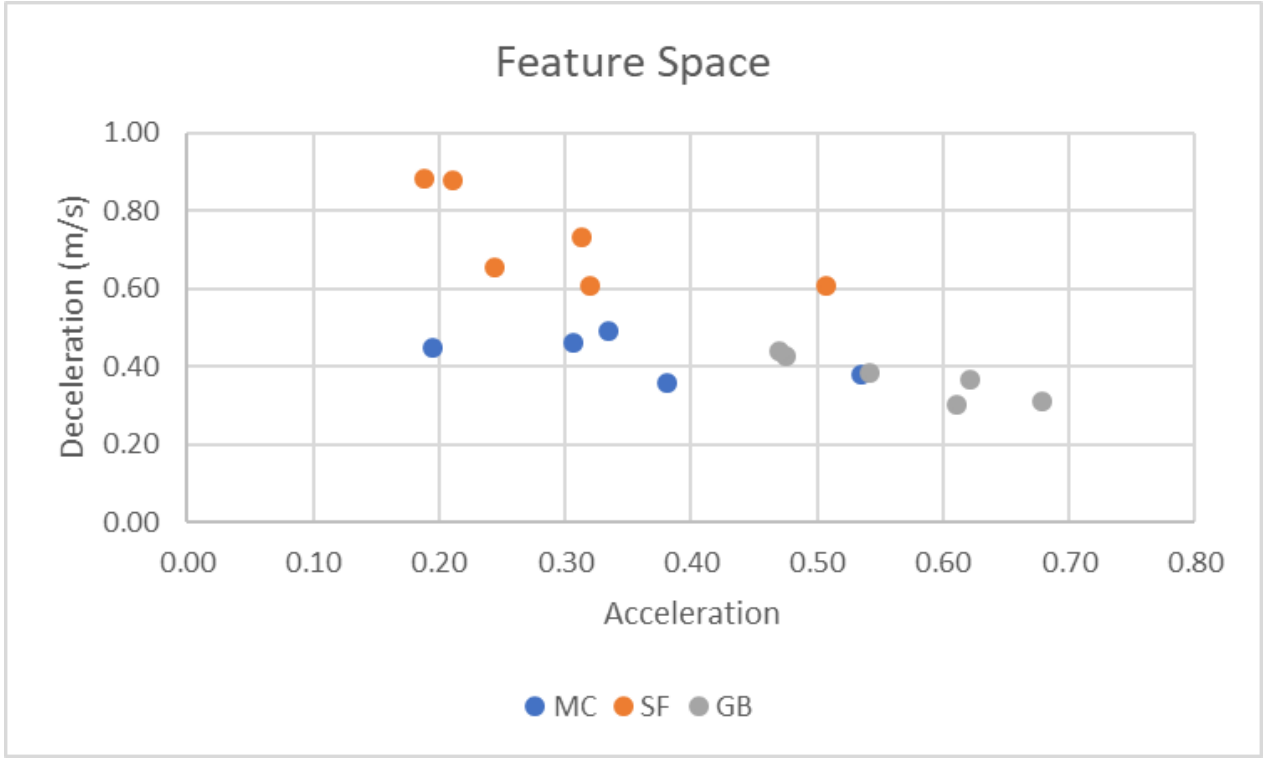


NAÏVE BAYES CLASSIFIER

We wish to create a system that would detect which elevator we are riding in based on:

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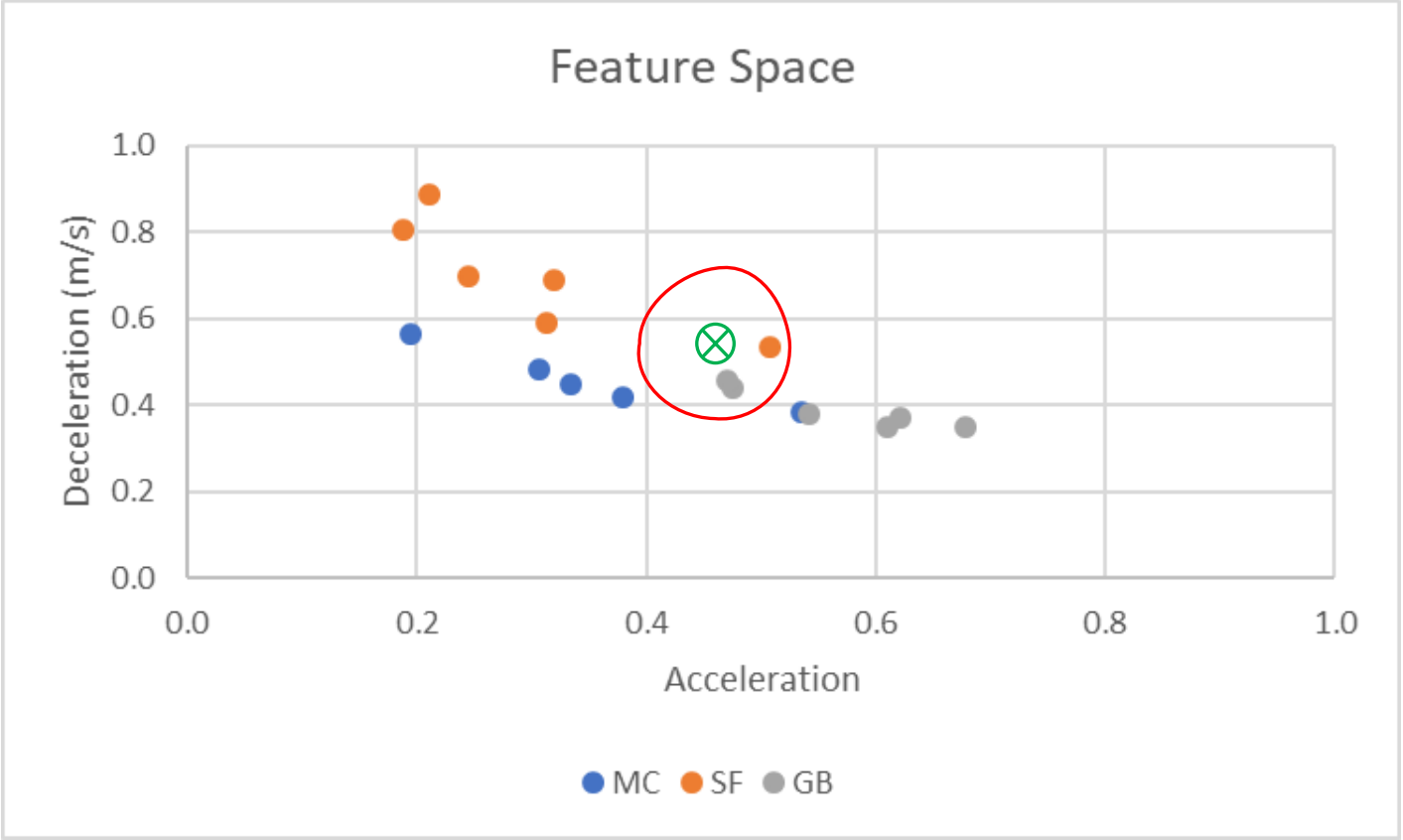
NAÏVE BAYES CLASSIFIER

Given a new data point ⊗,which elevator are we in?

If we use KNN with K=1, the elevator classified will be SF.

K=3 → elevator is GB

How can we use Bayes Theorem?

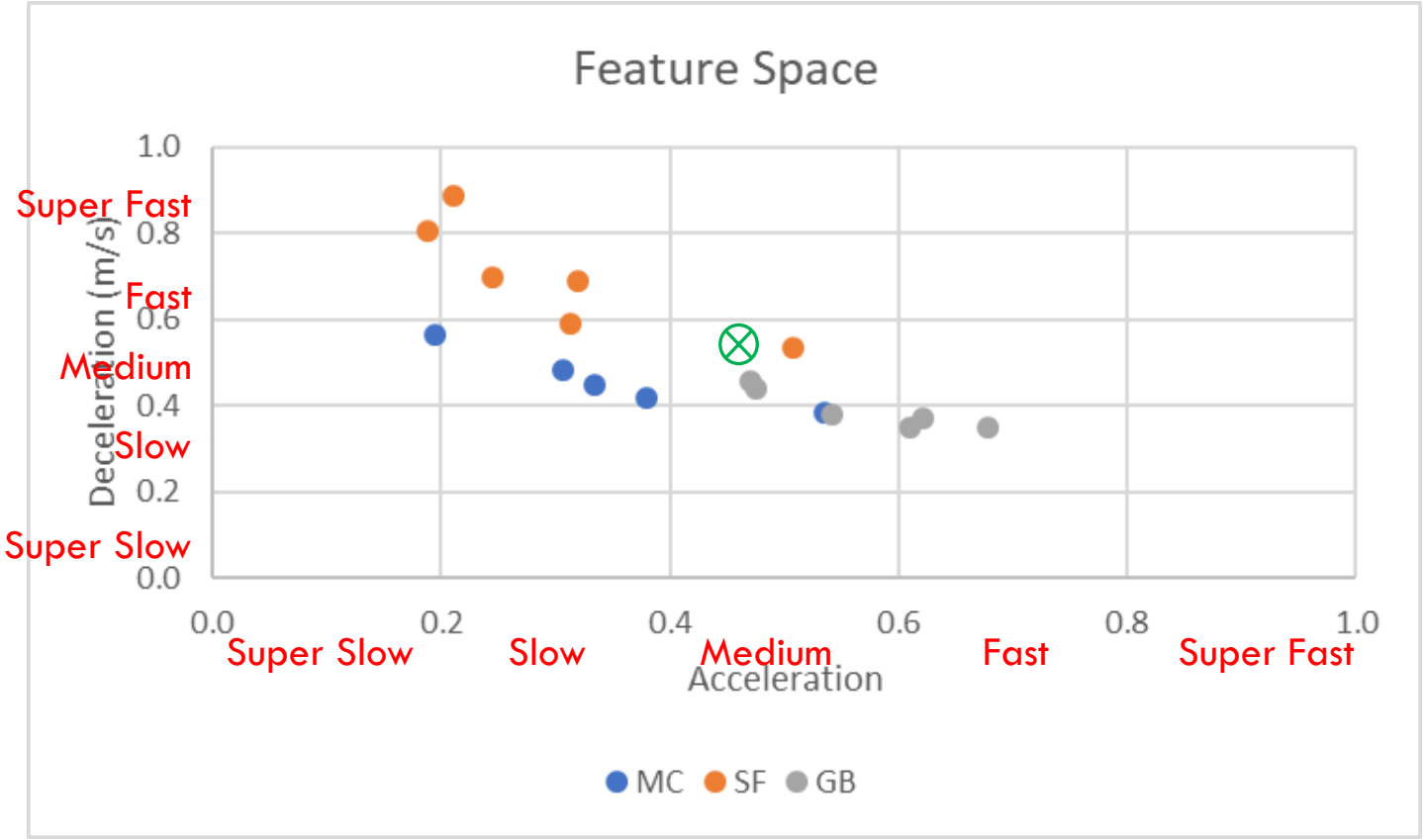


NAÏVE BAYES CLASSIFIER

While Bayes theorem can be used on continuous measurements – it will be simpler to discretize the data set.

In this example the data is discretized by 0.2 steps.

While labels are not required, they are used in this example for clarification



NAÏVE BAYES CLASSIFIER

Data point is:

Acceleration = Medium

Deceleration = Medium

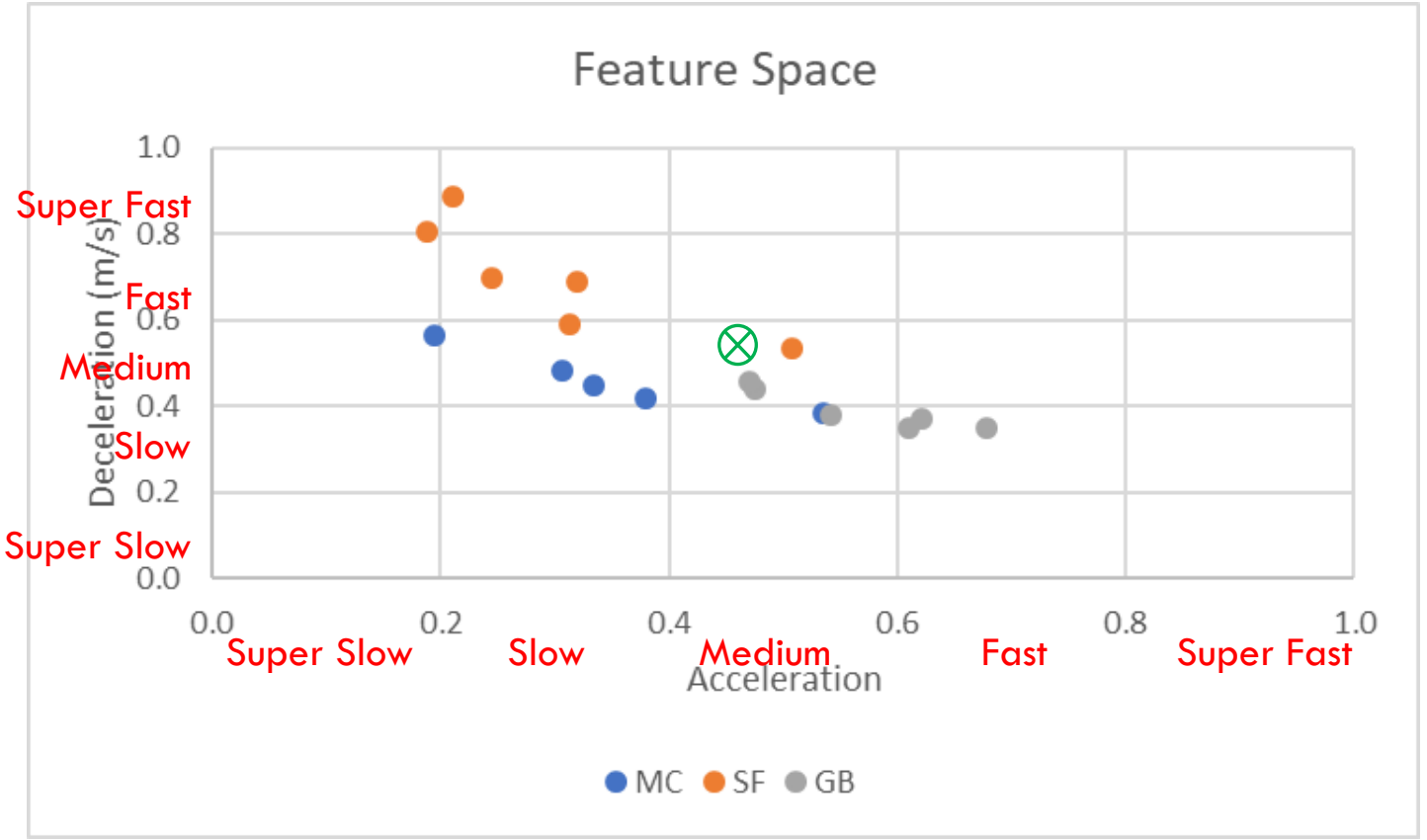
To be able to classify we need to calculate:

$P(X=MC \mid AC=M \cap DC=M)$

$P(X=SF \mid AC=M \cap DC=M)$

$P(X=GB \mid AC=M \cap DC=M)$

The one with greatest probability will be our selected elevator.



NAÏVE BAYES CLASSIFIER

Probability of multiple conditions is the multiplication of probability of each condition individually :

$P(X=MC|AC=M \cap DC=F) = P(X=MC|AC=M)*P(X=MC|DC=F)$

$$P(X=MC|AC=M) = \frac{P(AC=M|X=MC)*P(MC)}{P(AC=M)}$$

Assume that all elevators are visited the same number of times.

$P(MC)= 1/3$

$P(AC=M|X=MC)=?$

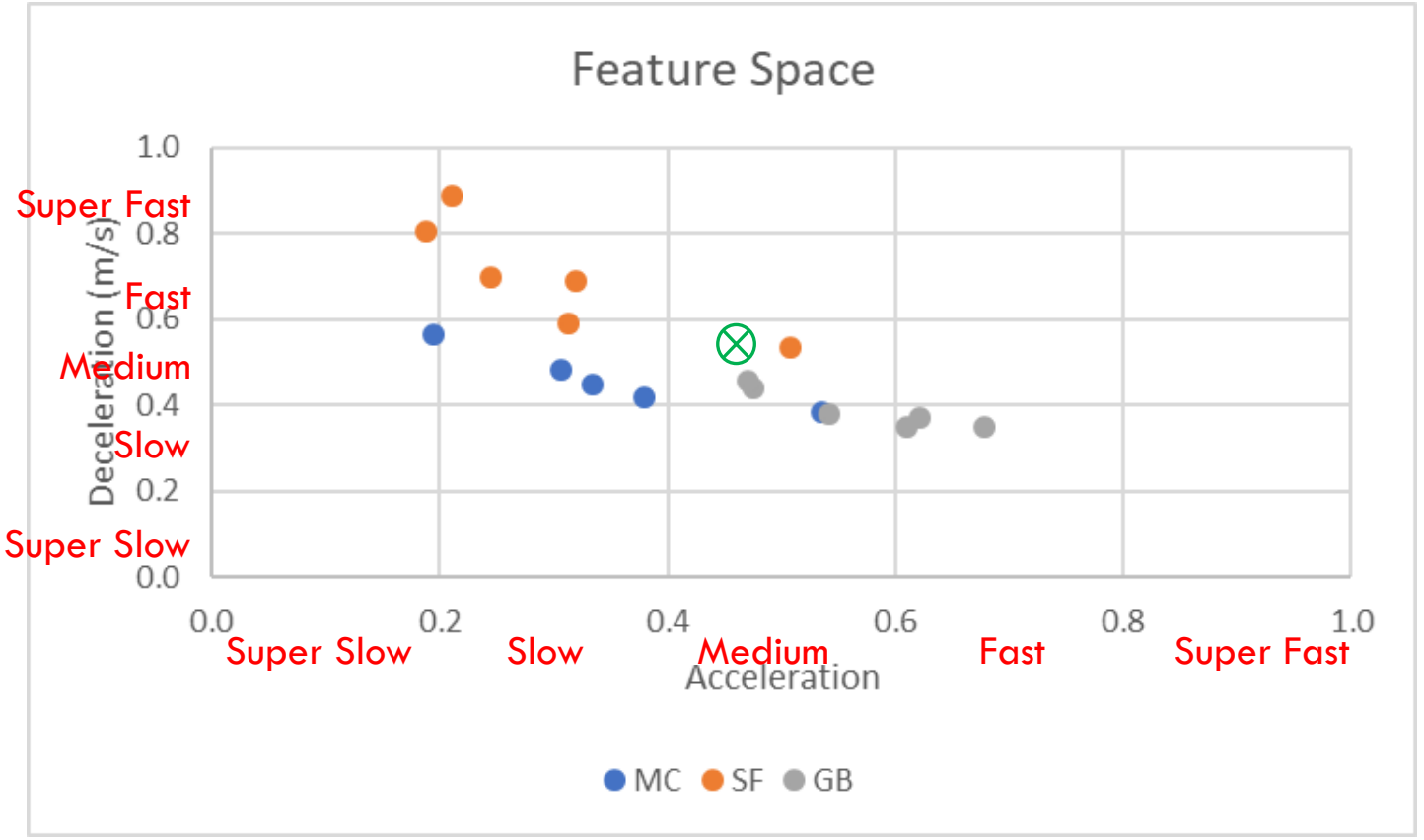
NAÏVE BAYES CLASSIFIER

P(AC=M | X=MC)=

P(AC=M)=

$$P(X=MC \mid AC=M) = \frac{P(AC=M \mid X=MC) * P(MC)}{P(AC=M)}$$

P(X=MC | AC=M) =



NAÏVE BAYES CLASSIFIER

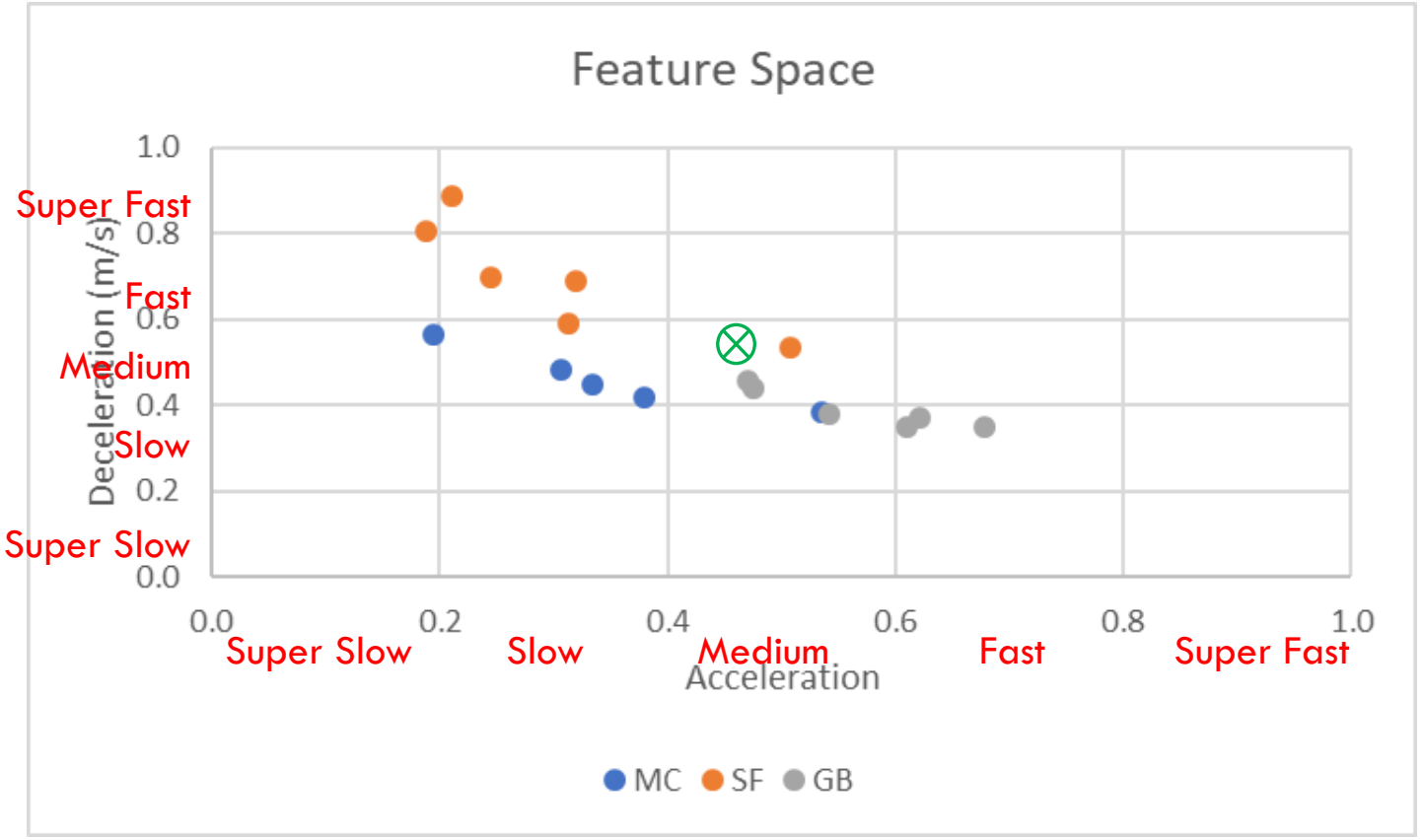
P(DC=M|X=MC)=

P(DC=M)=

$$P(X=MC|DC=M) = \frac{P(DC=M|X=MC)*P(MC)}{P(DC=M)}$$

P(X=MC|DC=M) =

P(X=MC|AC=M ∩ DC=M)=



NAÏVE BAYES CLASSIFIER

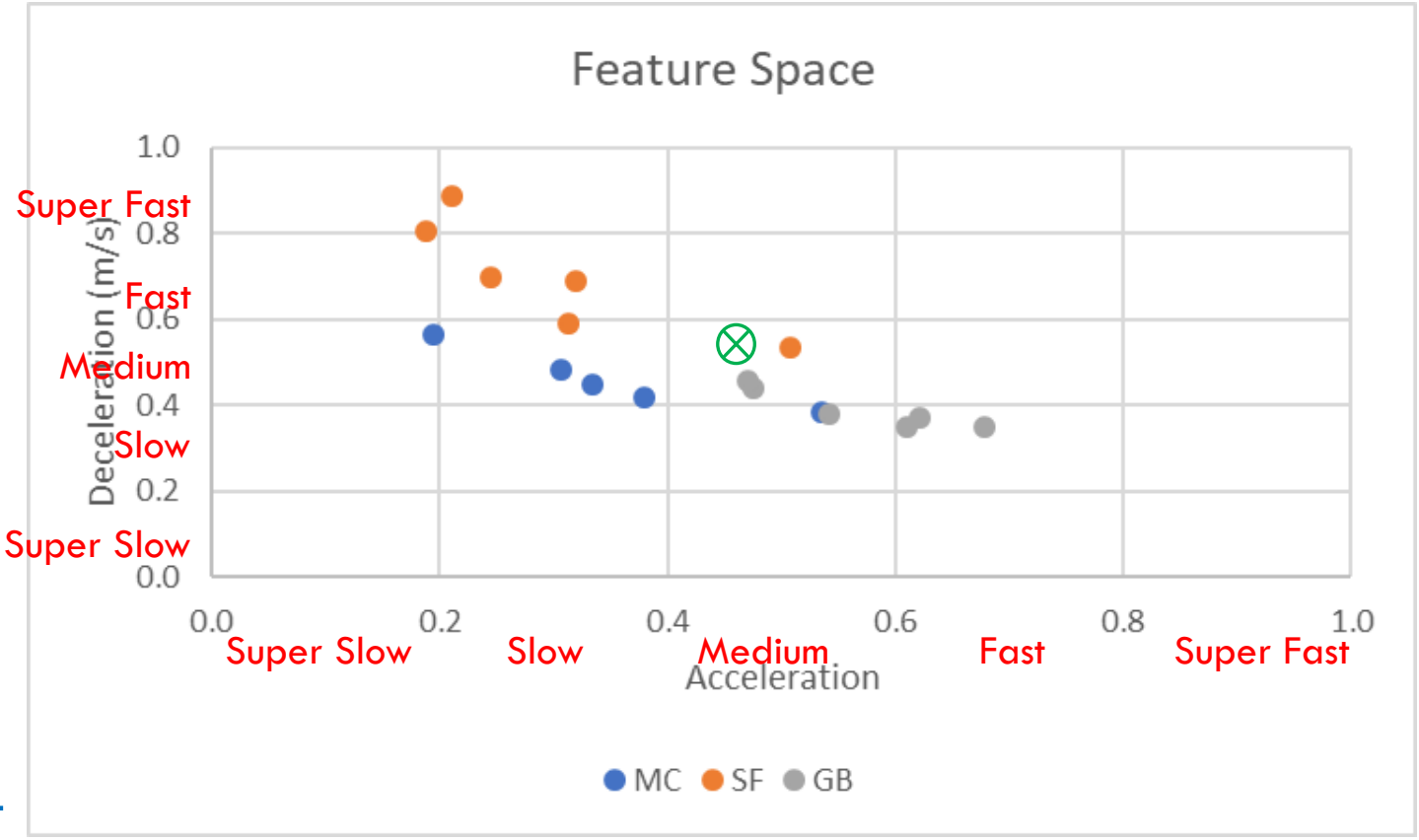
P(AC=M | X=SF)=1/6

P(AC=M)=5/17

$$P(X=SF \mid AC=M) = \frac{P(DC=M \mid X=SF)*P(SF)}{P(DC=M)}$$

P(X=SF | AC=M) = 0.189

P(X=SF | AC=M ∩ DC=M)=
0.189*0.236=0.044



NAÏVE BAYES CLASSIFIER

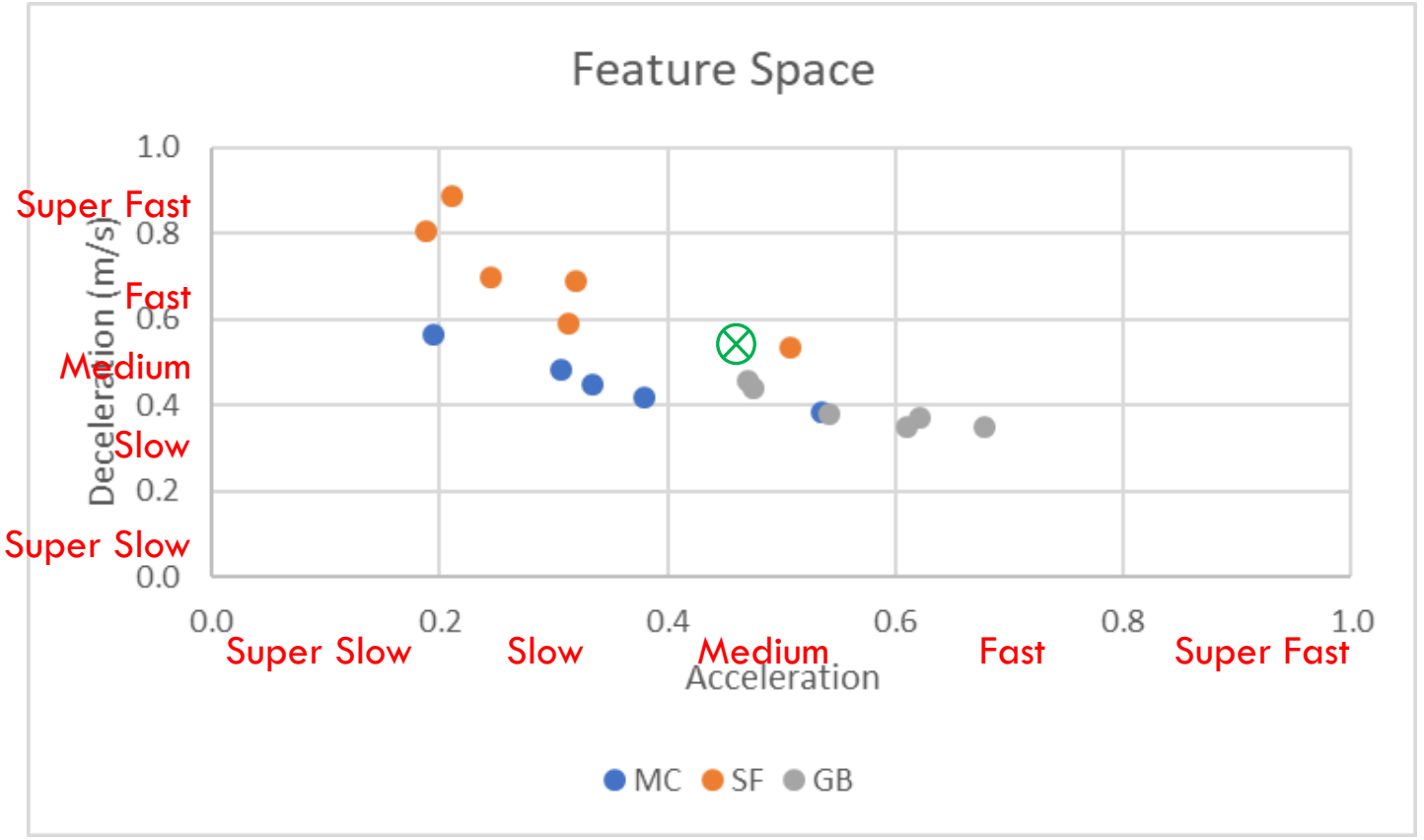
P(AC=M | X=GB)=3/6

P(AC=M)=5/17

$$P(X=GB \mid AC=M) = \frac{P(DC=M \mid X=GB)*P(MC)}{P(DC=M)}$$

P(X=GB | AC=M) = 0.566

P(X=GB | AC=M ∩ DC=M)=
0.56*0.236=0.132



NAÏVE BAYES CLASSIFIER

P(X=GB | AC=M ∩ DC=M)= 0.132

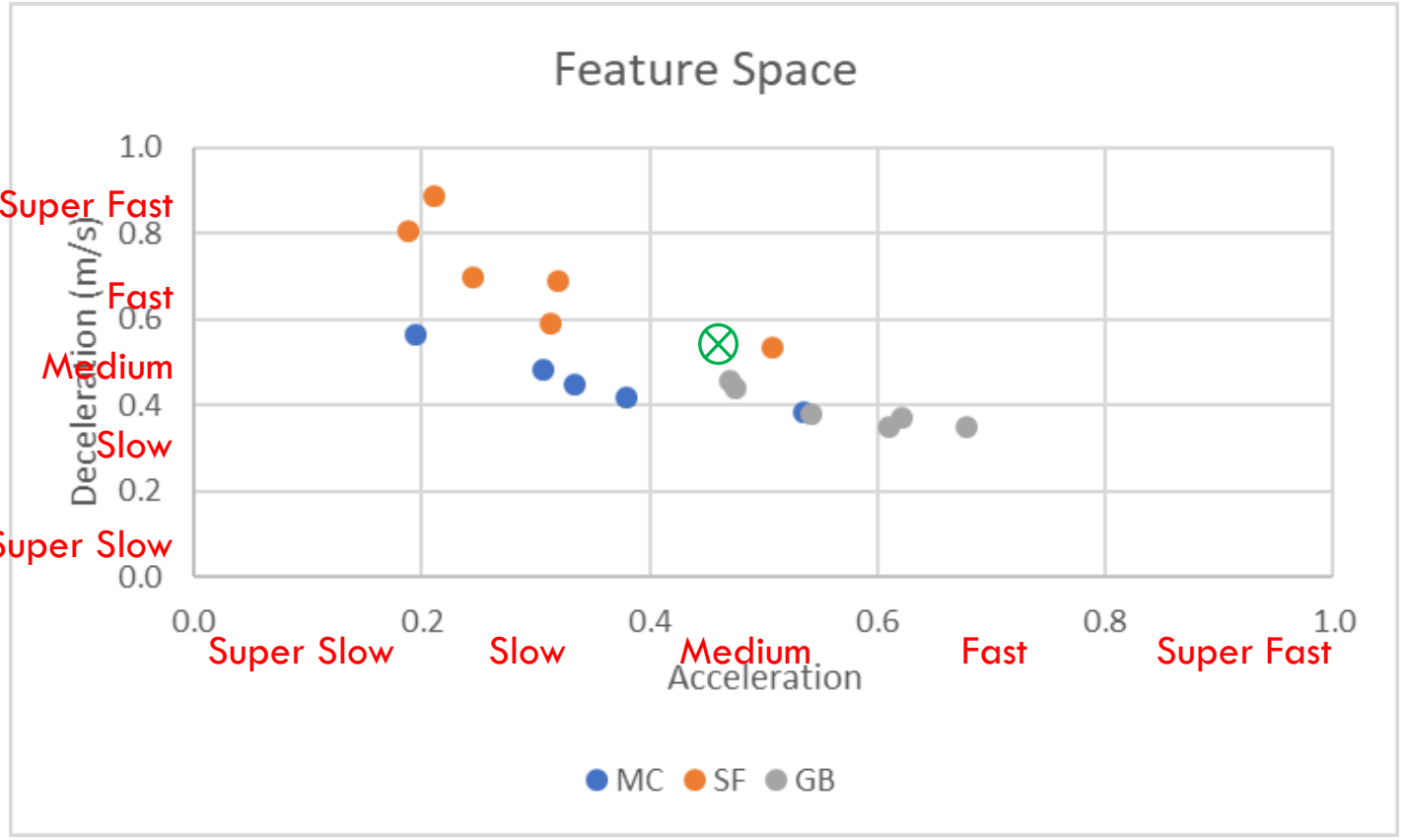
P(X=SF | AC=M ∩ DC=M)= 0.044

P(X=MC | AC=M ∩ DC=M)= 0.128

Therefore we can declare the elevator to be GB.

Why don't the probabilities add up to 100%?

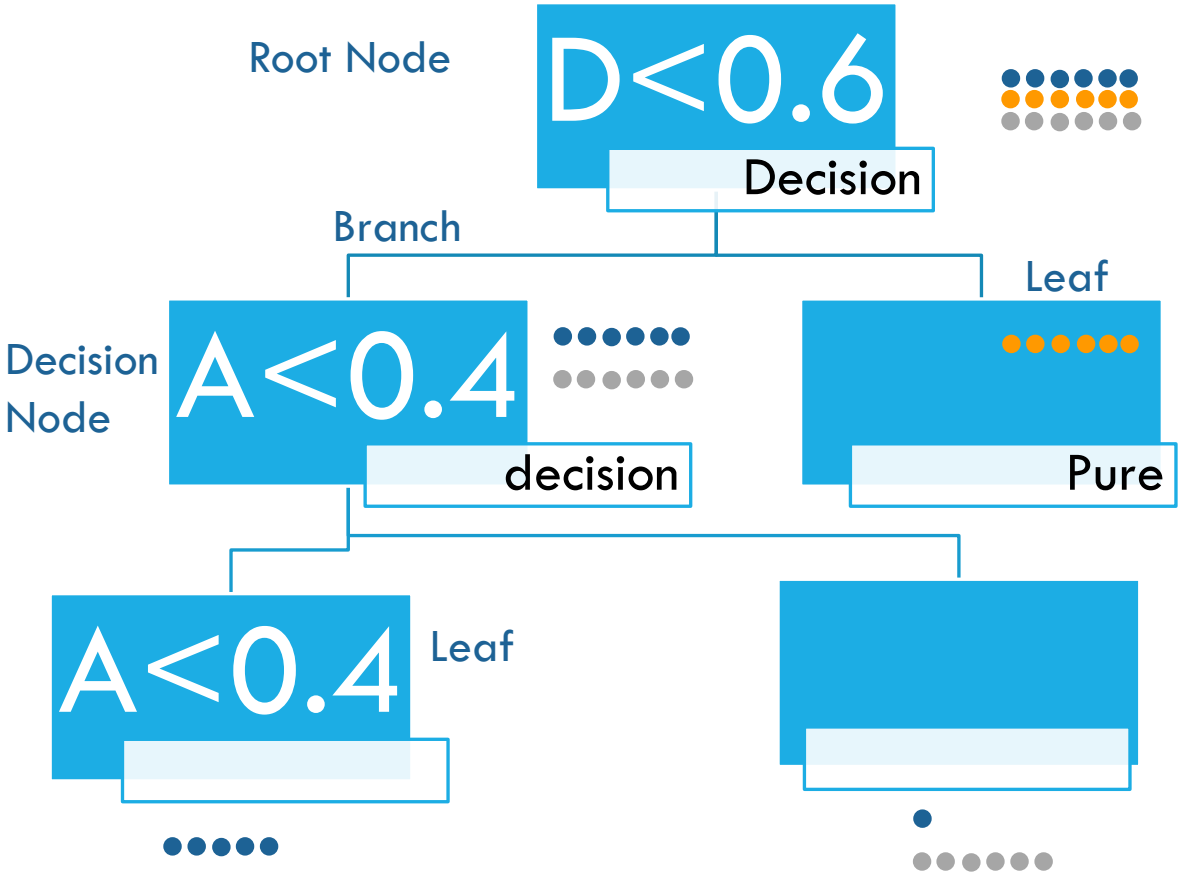
What is the confidence of your answer?

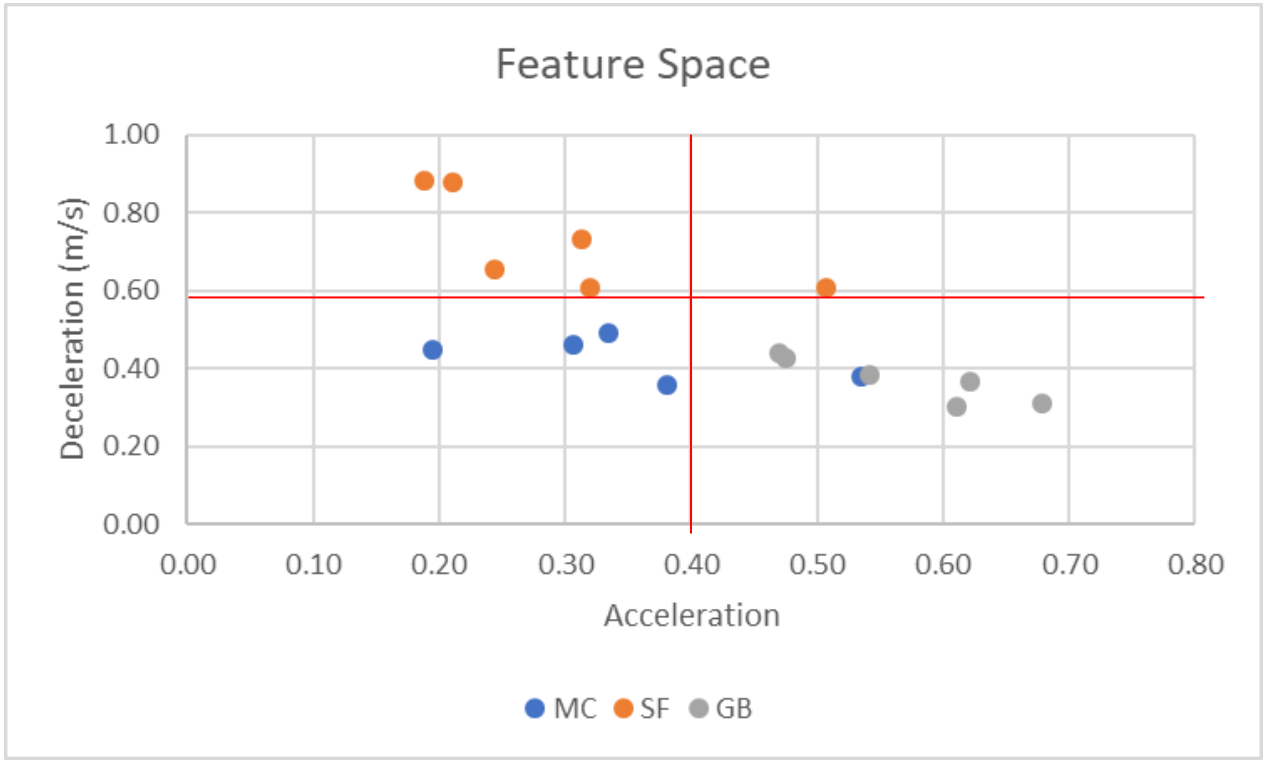


DECISION TREE

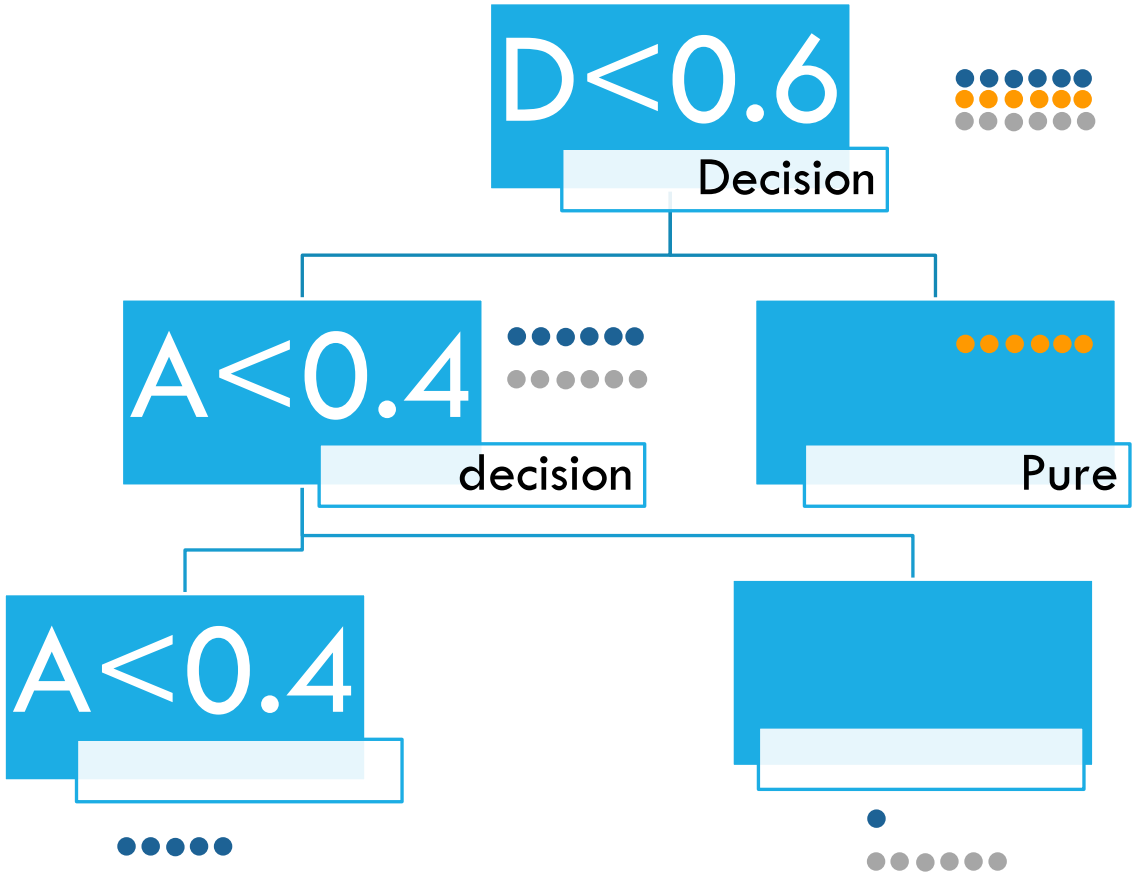
- A decision tree is a flowchart-like structure where an internal node represents a feature (or attribute), a branch represents a decision rule, and each leaf node represents the outcome.
- The decision variables are trained in a greedy method to maximize information gain

DECISION TREE CLASSIFICATION



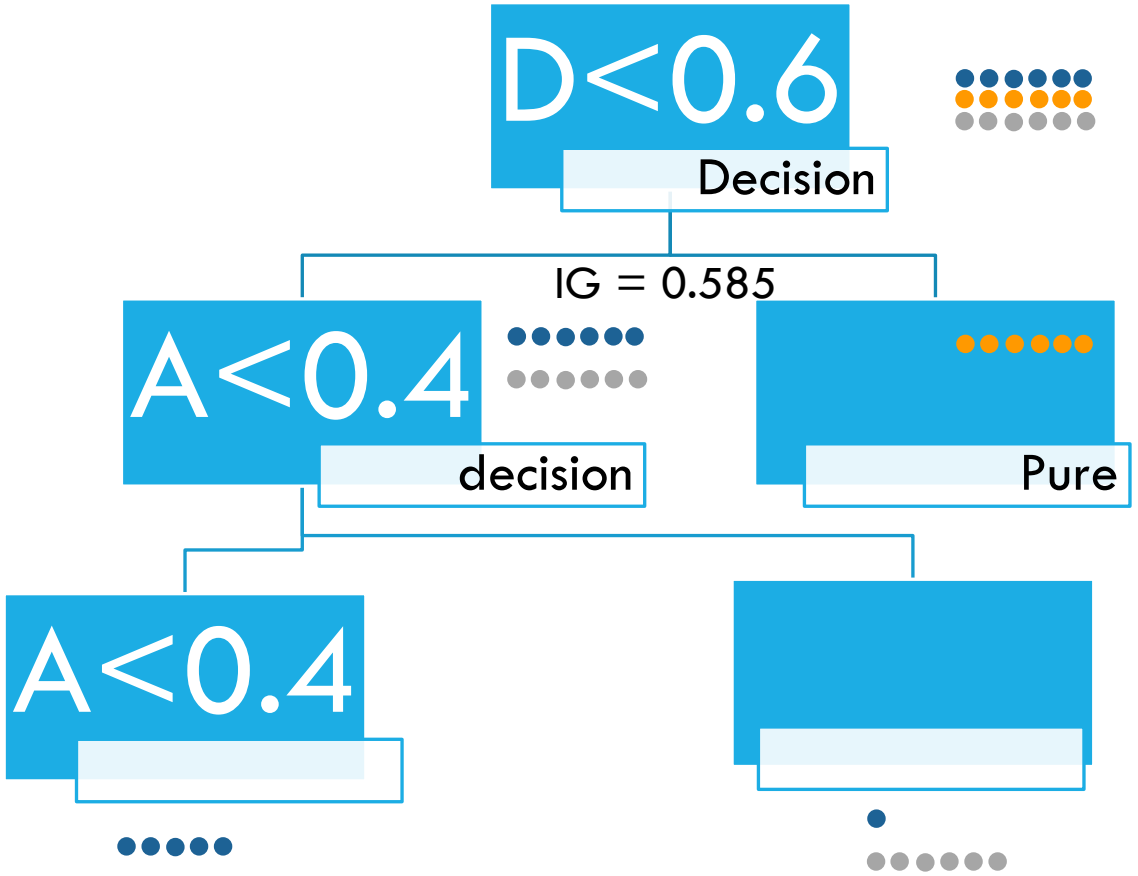


DECISION TREE CLASSIFICATION



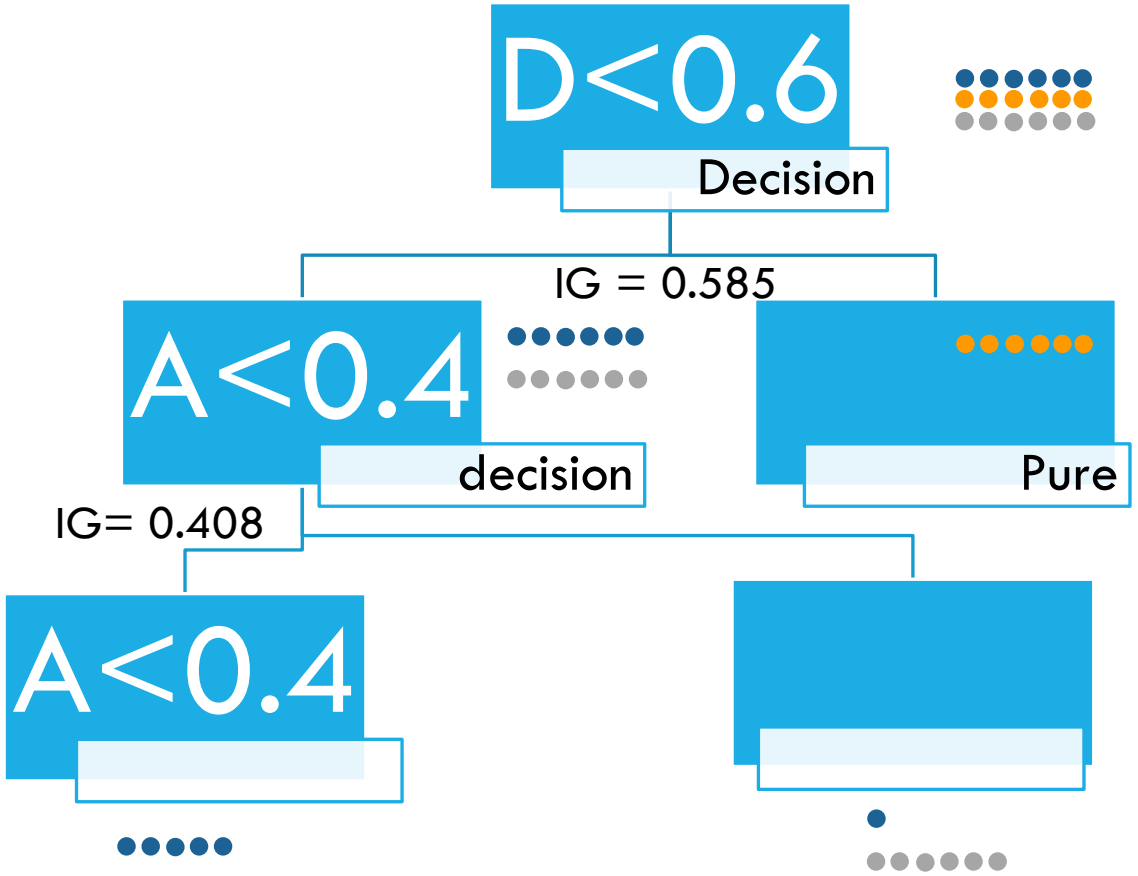
- Entropy – a form of uncertainty in your decision success.
- Entropy = $\sum$ $-P_i * \log(P_i)$
- $E_1 = -0.333*\log_2(0.333)-0.333*\log_2(0.333)-0.333*\log_2(0.333)$
- $E_1 = 1.585$
- Max entropy depends on the number of outcomes.
- Max $E=\text{Log2}(n)=\log_2(3)=1.585$

DECISION TREE CLASSIFICATION



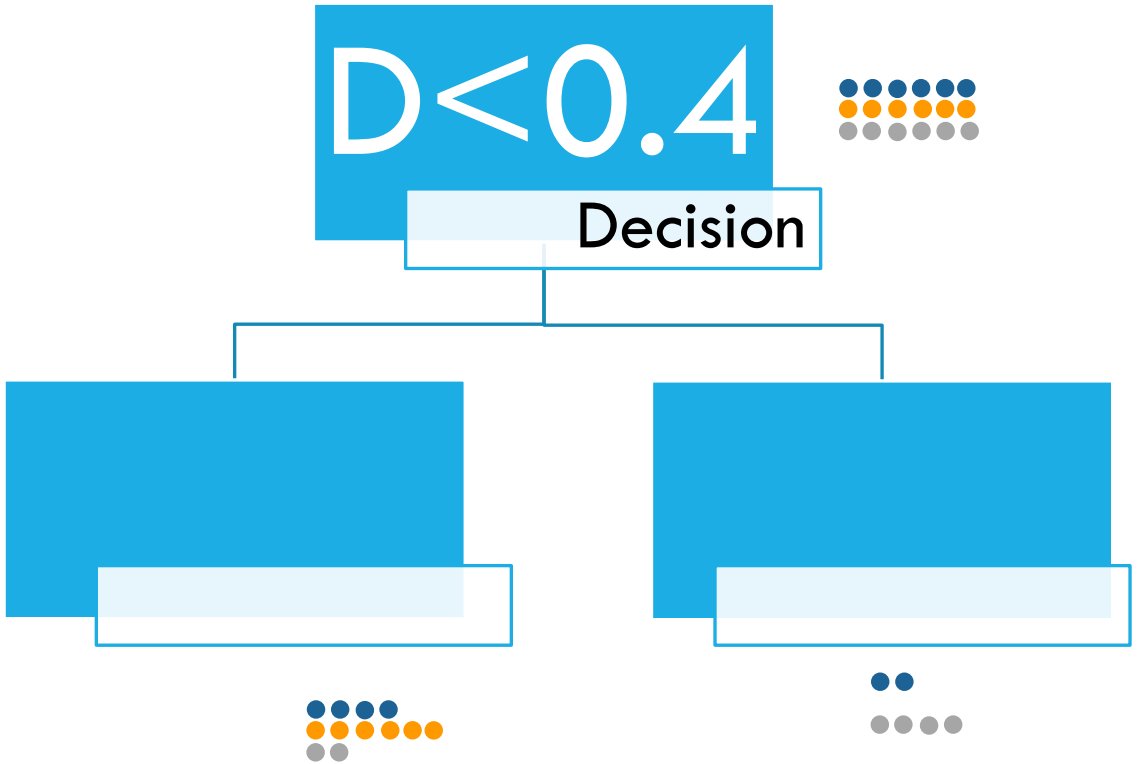
- Entropy = $\sum$ -P_i* log(P_i)
- E_2= -0.5*$\log_2$(0.5)-0.5*$\log_2$ (0.5)
- E_2 =1
- E_3= 1*$\log_2$(1)=0
- Information gain = E_1 –(E_2 W_2 +E_3 W_3)=1.585-1*(12/18) – 0*(6/18) =0.918

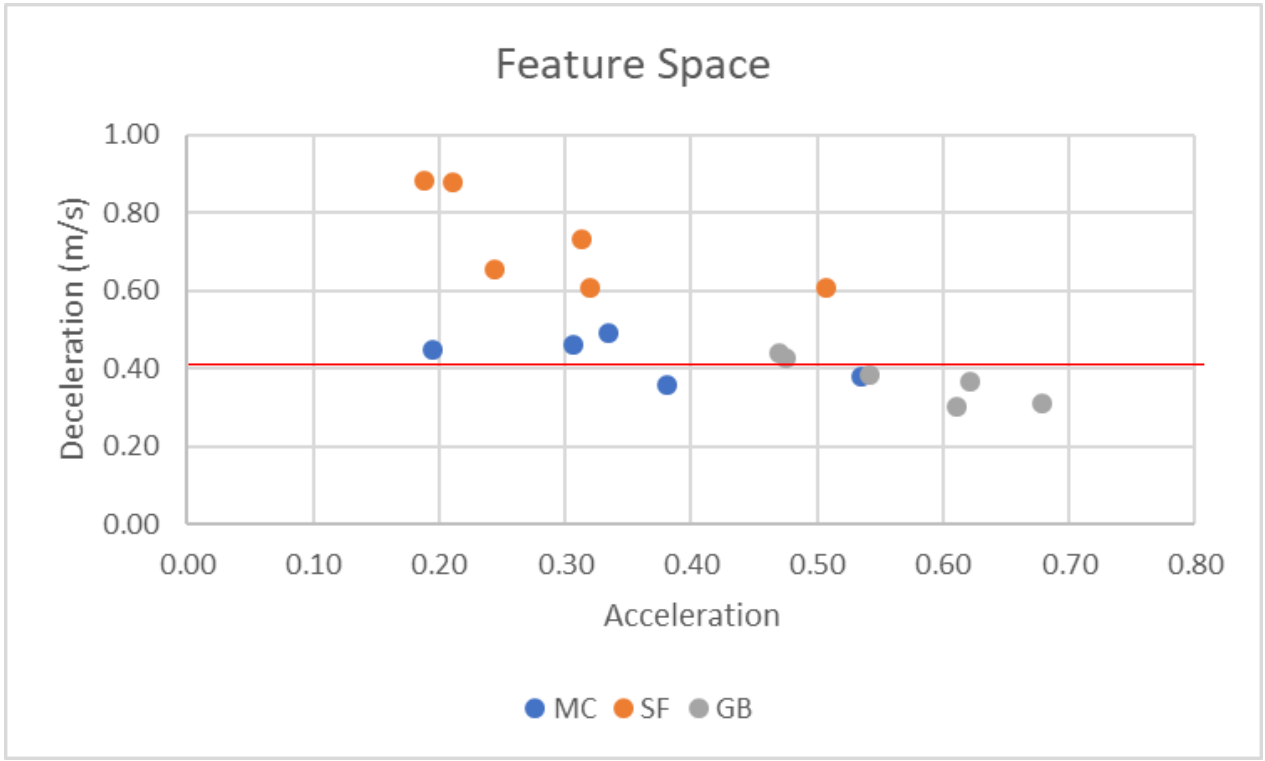
DECISION TREE CLASSIFICATION



- E_4= 1*$\log_2(1)$=0
- E_5= -1/7*$\log_2(1/7)$-6/7*$\log_2(6/7)$ =0.592
- Information gain =1-0.592(7/12) = 0.655

DECISION TREE CLASSIFICATION



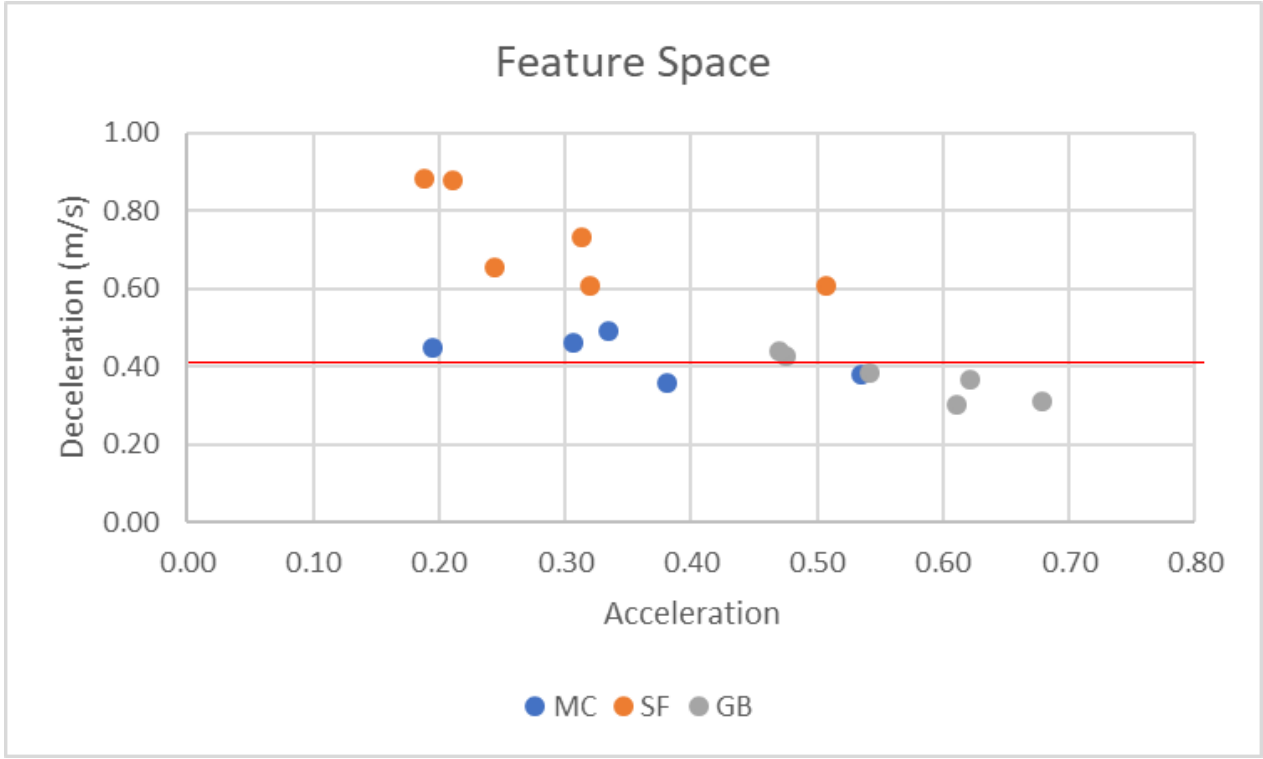


What is the information gain in this example?

DECISION TREE CLASSIFICATION

Training

- Use of a greedy method to maximize Information gain.
- Greedy method will find a solution but not guarantee an optimum one.
- What is the problem with decision tree?



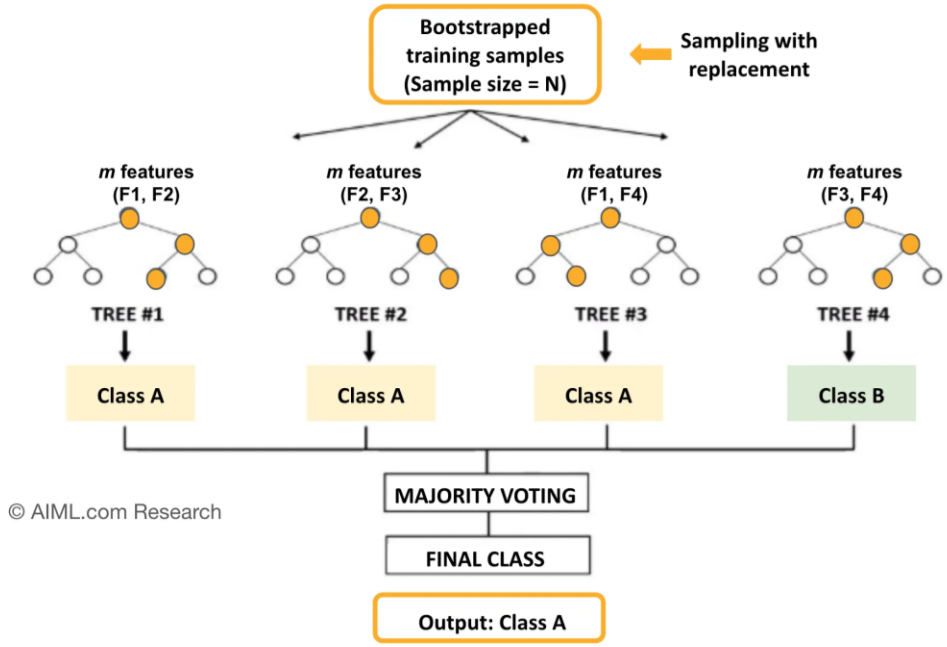
RANDOM FOREST ALGORITHM

- Random Forest is an extension of decision tree with 2 major changes:

1. Bootstrapping – creating multiple data sets of the original data set by duplicating training points (randomly).
2. Selecting a subset of features (randomly) to train a decision tree on each sample set.
3. Unknown data points are passed through all trees and a vote is taken.

Training Data
(Sample size, N= 6, No. of features, F = 4)

F1	F2	F3	F4	Y
2.1	0	400	-9	A
3.0	1	890	-42	B
2.2	1	929	0	B
4.0	0	324	-23	A
3.5	1	333	-15	A
6.0	0	215	-9	A



BOOTSTRAPPING

- A **bootstrap sample** is made by randomly selecting items **with replacement** — meaning some data points may appear more than once, others not at all.

Training Data
(Sample size, N= 6, No. of features, F = 4)

F1	F2	F3	F4	Y
2.1	0	400	-9	A
3.0	1	890	-42	B
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3.5	1	333	-15	A
6.0	0	215	-9	A

F1	F2	F3	F4	Y
2.1	0	400	-9	A
4.5	0	324	-23	A
3.5	1	333	-15	A
6.0	0	215	-9	A
3.5	1	333	-15	A
4.5	0	324	-23	A

Bootstrap sample 1

F1	F2	F3	F4	Y
2.1	0	400	-9	A
4.5	0	324	-23	A
3.5	1	333	-15	A
6.0	0	215	-9	A
2.1	0	400	-9	A
3.0	1	890	-42	B

Bootstrap sample 1

RANDOM FOREST: REAL OCCUPANCY DATA

The data

- Office room, one reading per minute (UCI Occupancy dataset).
- **Features:** temperature, humidity, light, CO2, humidity ratio. **Label:** occupied or empty.
- **Train:** 8,143 readings (Feb 4–10). **Test:** 12,417 from other days.

The forest

- 100 trees, each trained on its own bootstrap sample.
- Each split may use only 2 of the 5 features, chosen at random.
- New readings go through all 100 trees; the majority vote wins.

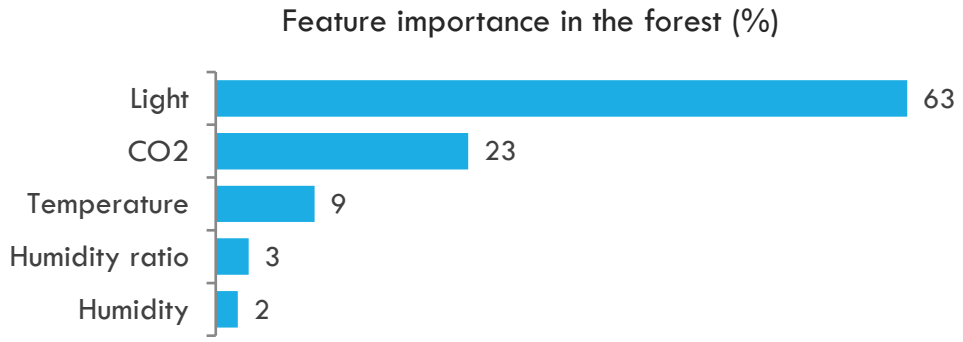
Lessons

- Averaging 100 trees cut the errors almost in half, and the result barely changes with the random seed.
- Out-of-bag accuracy was 99.4%, but 97.0% on new days. Readings one minute apart are near-copies, so OOB is optimistic.
- **Baseline check:** the rule light > 365 lux alone scores 99.0%.

Results on the 12,417 test readings

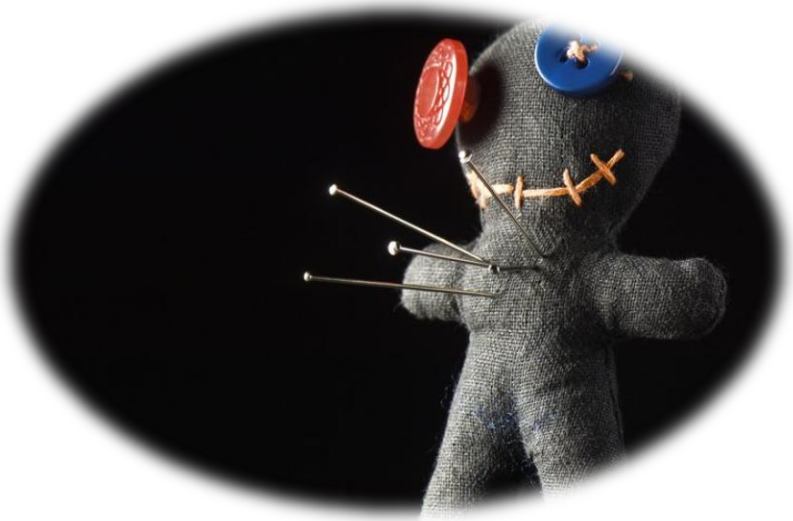
Model	Accuracy	Errors
Single decision tree	94.0%	743
Random forest, 100 trees	97.0%	399

Single tree: 92.9% to 95.4% depending on the random seed.



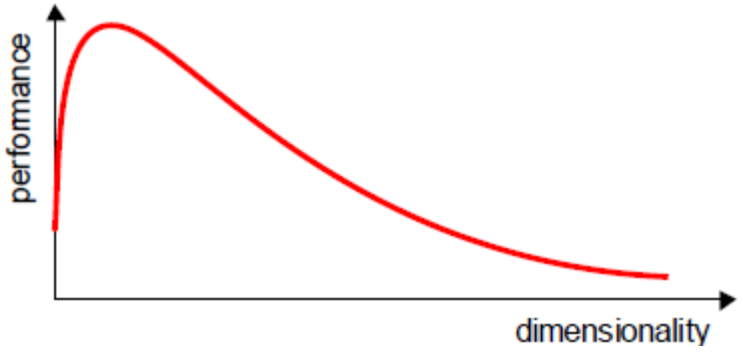
One vote, Feb 2 at 14:22: light 494 lux, CO2 775 ppm, 23.7 °C. **65 of 100 trees vote occupied** → occupied (correct), with 65% of the votes.

DIMENSIONALITY REDUCTION



Curse of dimensionality

- The complexity of the classification problem is proportional to the number of categories multiplied by the number of features.
- The number of training data set required for accurate estimation increases exponentially with the number of feature.



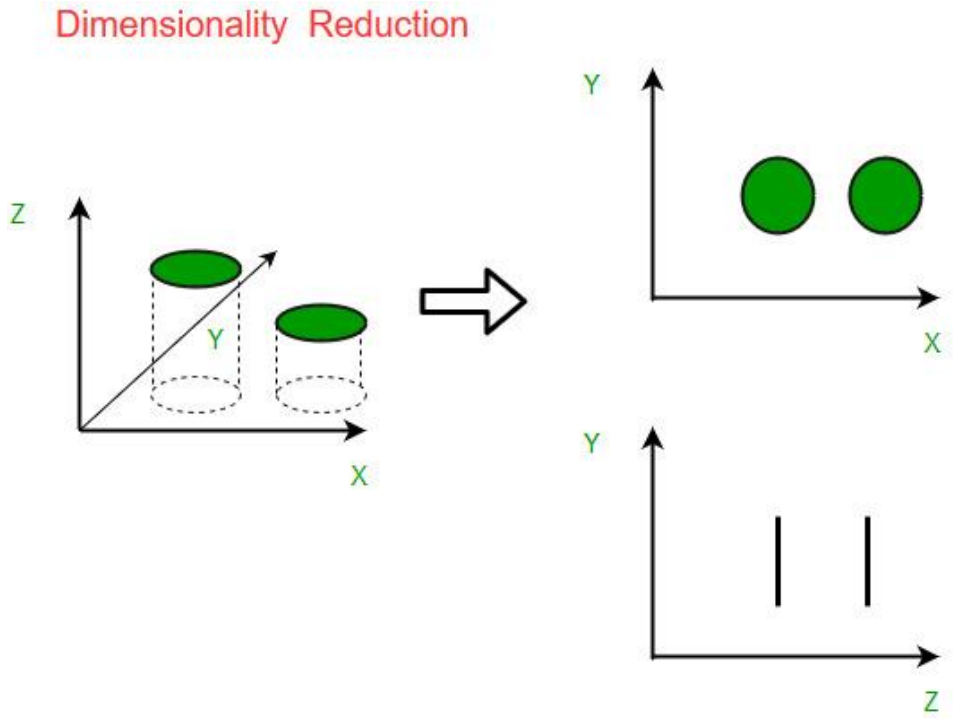
DIMENSIONALITY REDUCTION

Dimensionality reduction

1. Feature Selection: choosing a subset of all the features

2. Feature Extraction: creating new features by combining existing ones

- These two techniques can be used separately or in conjunction to reduce dimensionality.



DIMENSIONALITY REDUCTION

Feature Selection

A feature selection algorithm is a search technique for proposing new feature subsets.

Wrapper techniques:

- A subset of all features is selected.
- Classification is done on training data set
- Errors are calculated on the verification data set and used to score the subset.
- Subset with highest score (lowest error) is selected.

While most accurate this is a computationally expensive search method.

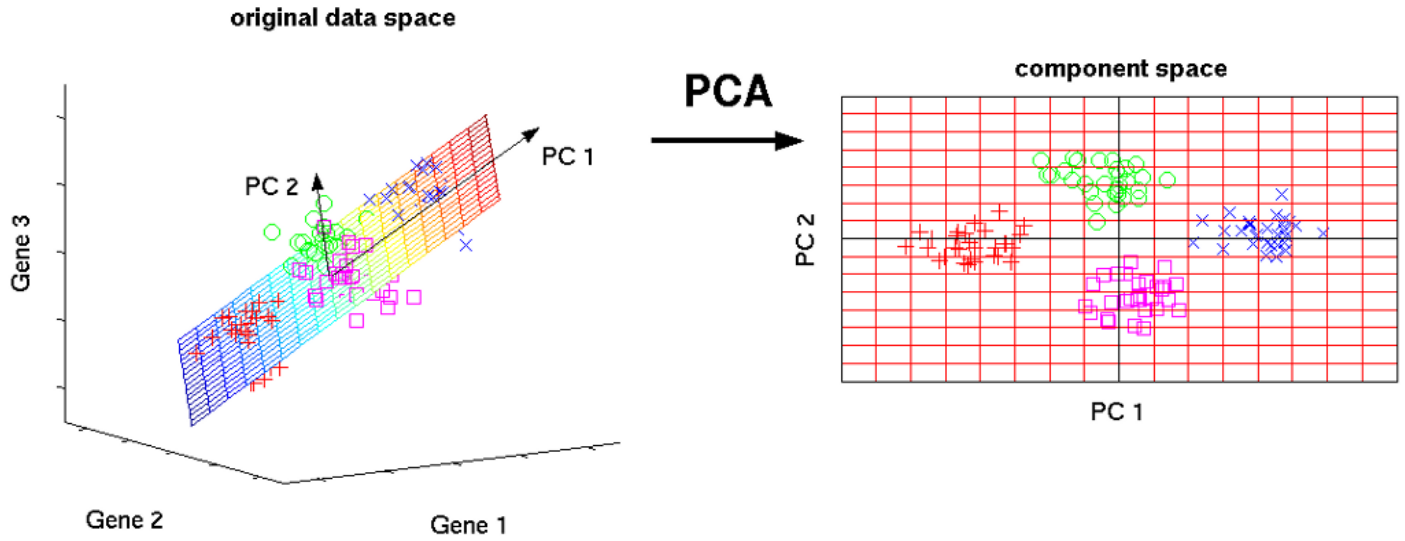
Other methods such as LASSO (least absolute shrinkage and selection operator) exist that provide feature selection and normalization.

DIMENSIONALITY REDUCTION

Feature Extraction/Feature Projection

The projection of high-dimensional space to a space of fewer dimensions can be accomplished using linear and non-linear transformation.

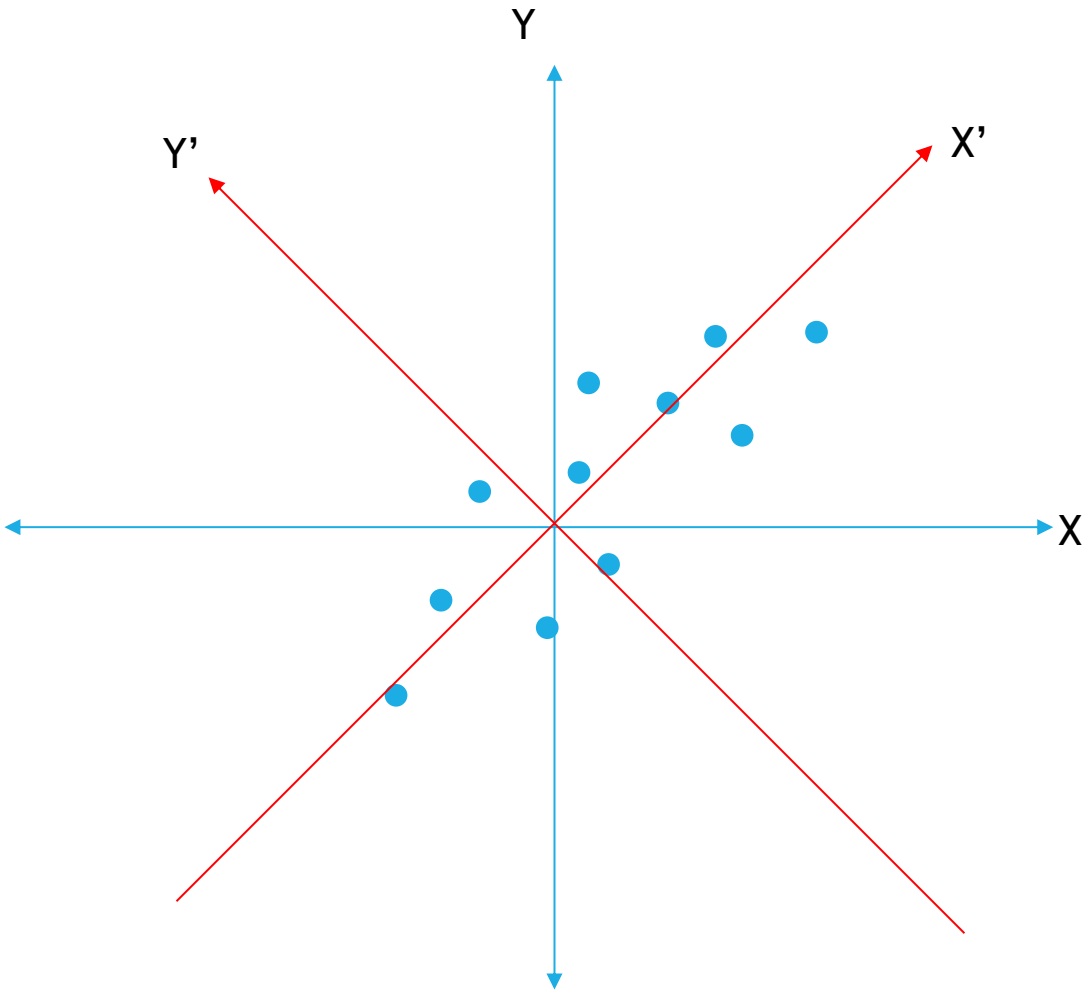
Principle Component Analysis (PCA) is the primary linear technique used for feature extraction.



DIMENSIONALITY REDUCTION

Principle Component Analysis (PCA)

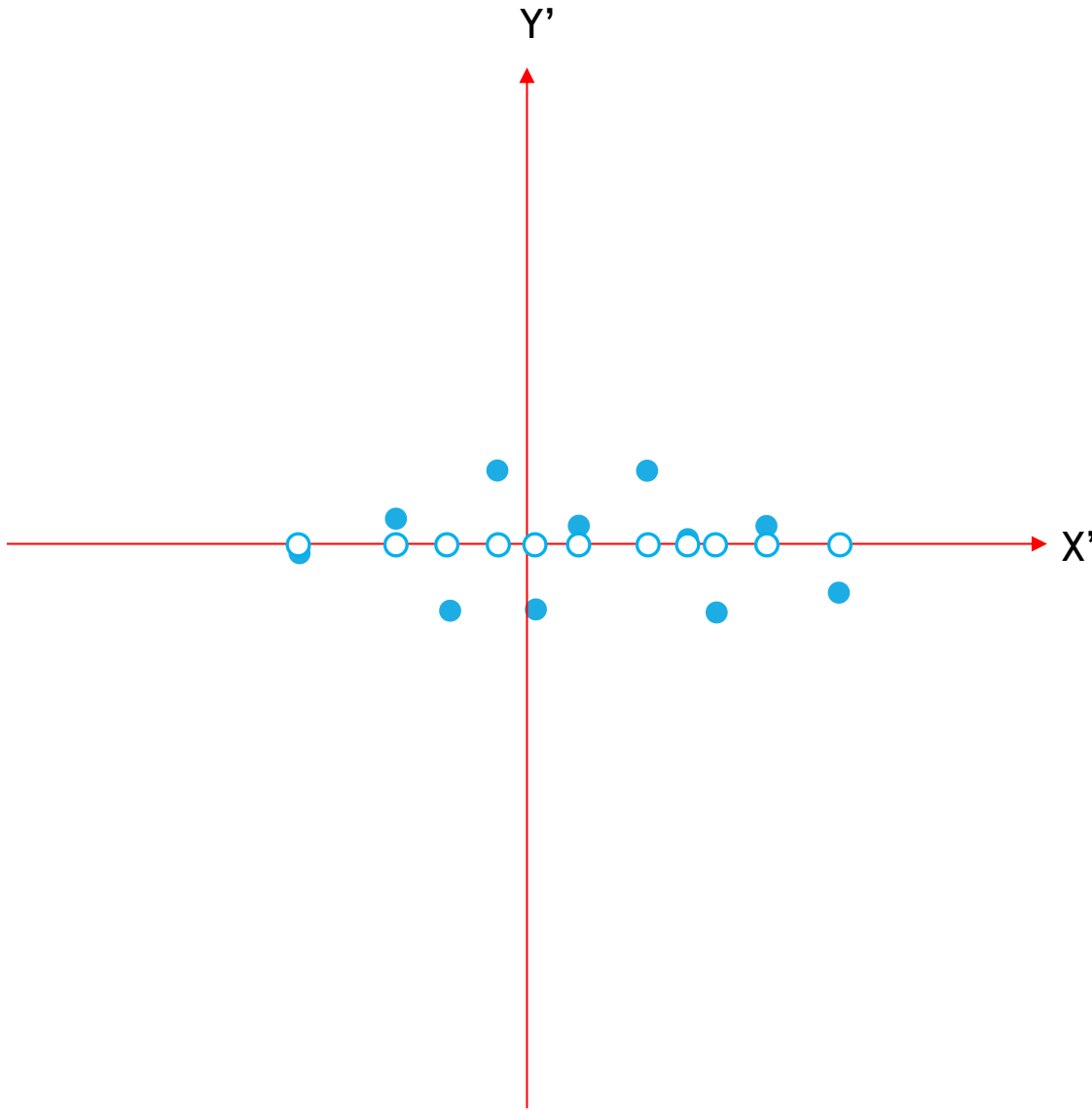
- Assume there are 2 features in our data set, X and Y.
- The data has nearly equal variance in both X and Y features.
- This means both are equally important to the classification model.
- If we define a new vector X' and Y':
  - Variance in X' is much more than Variance in Y'



DIMENSIONALITY REDUCTION

Principle Component Analysis (PCA)

- Since variance in Y' is much less than variance in X' we can classify the data set using only the X' feature.



DIMENSIONALITY REDUCTION

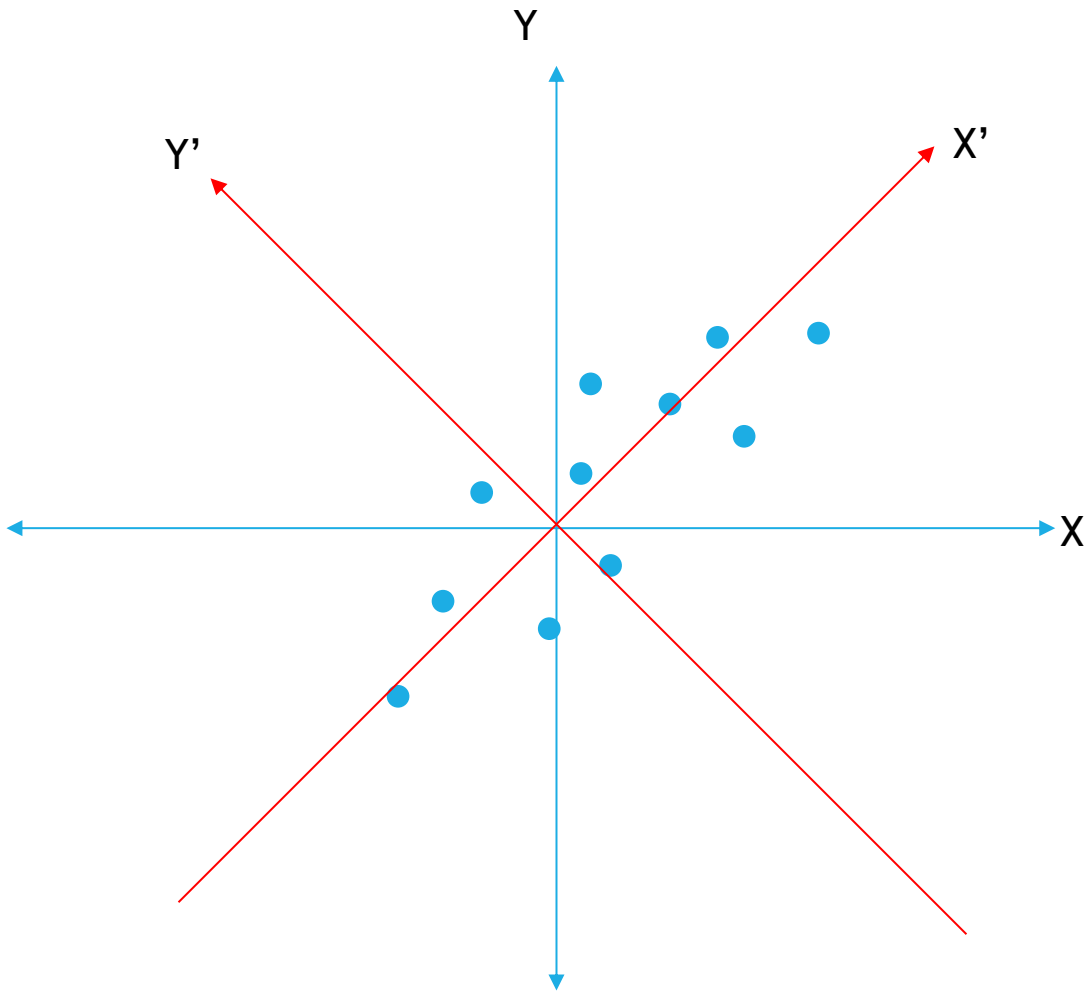
Principle Component Analysis (PCA)

1. Calculate variance for X and Y.
2. Calculate co-variance for X,Y
3. Create the Sigma (co-variance) Matrix

$$\Sigma=\begin{bmatrix} Var(X) & Cov(X,Y) \\ Cov(X,Y) & Var(Y) \end{bmatrix}$$

4. All data points can now be mapped into the new coordinates

$$\begin{bmatrix} X' \\ Y' \end{bmatrix} = \Sigma \begin{bmatrix} X \\ Y \end{bmatrix}$$

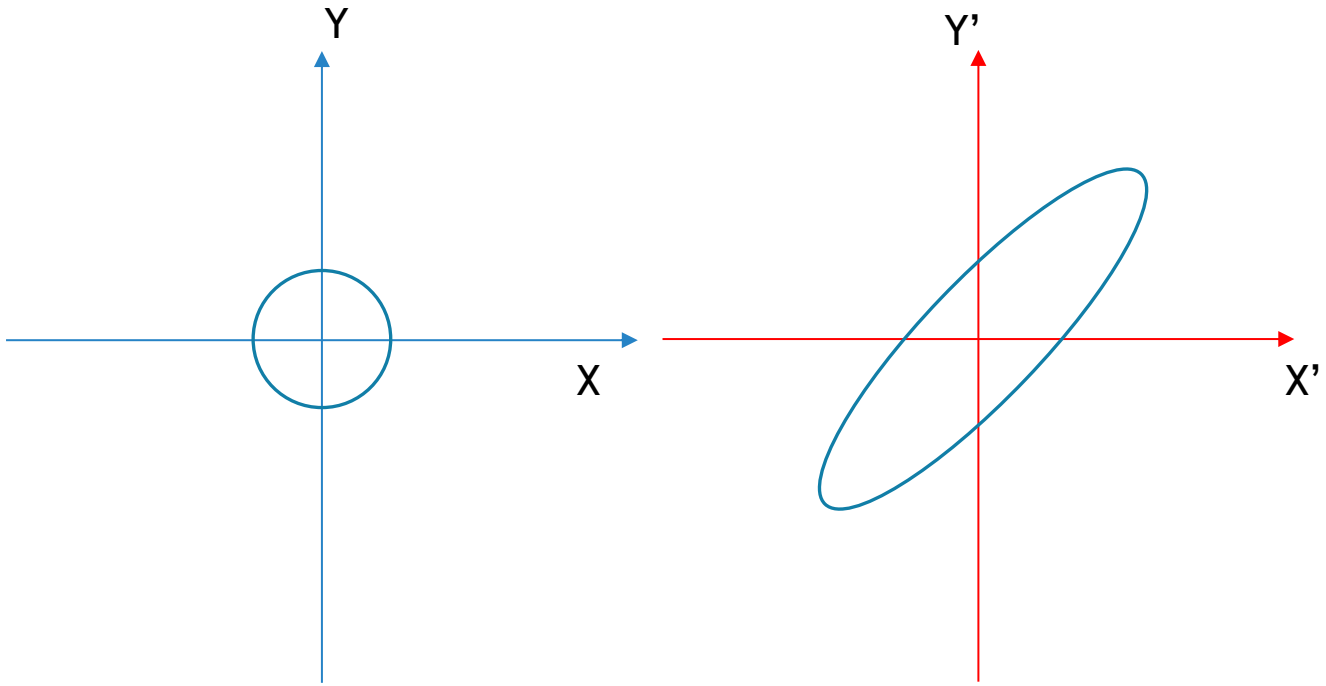


DIMENSIONALITY REDUCTION

Principle Component Analysis (PCA)

The transformation will result in shift and scaling of the data points.

The transformation can be described by the eigenvector and the eigenvalues.



DIMENSIONALITY REDUCTION

Principle Component Analysis (PCA)

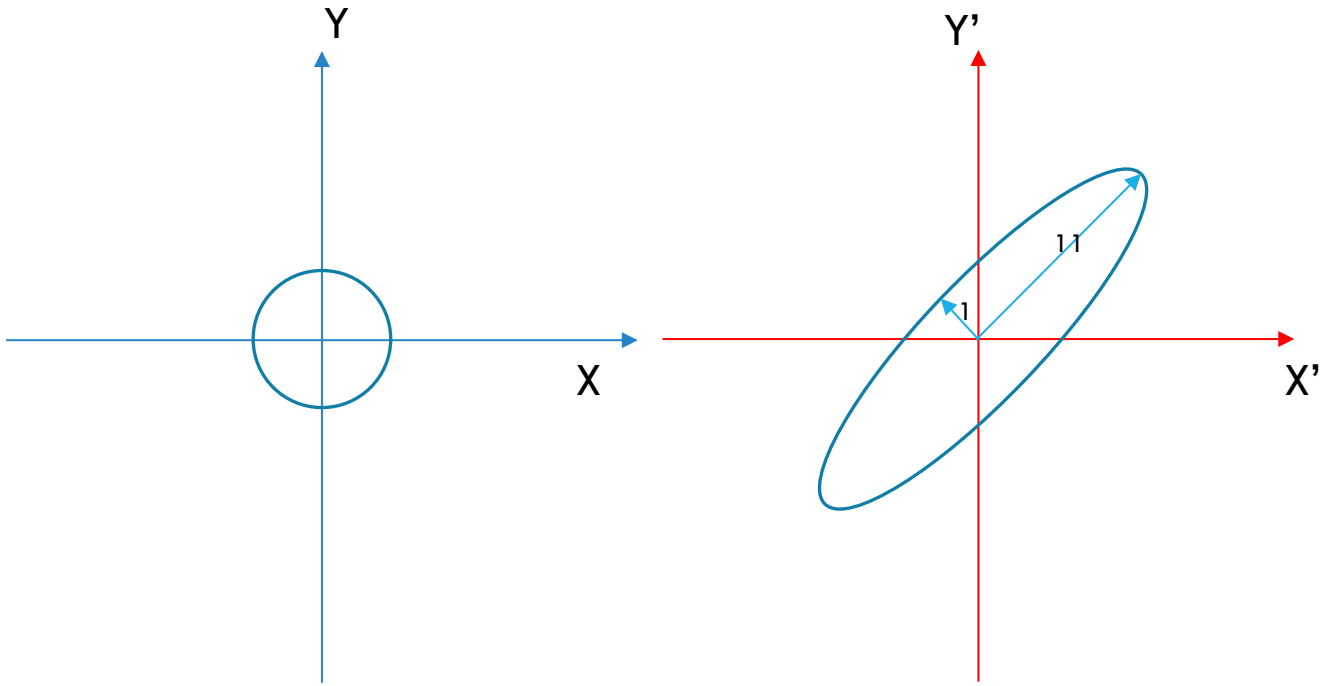
For example

$$\Sigma=\begin{bmatrix}9 & 4\\ 4 & 3\end{bmatrix}$$

Eigenvectors are [2,1] and [-1,2], and

Eigenvalues are 11 and 1 respectively.

For this example we may decide to remove Feature 2 since its eigenvalue is only 1.



SUMMARY

For any classification or regression analysis we must:

1. Collect training and verification data
2. Prepare data (scale and remove outlier if possible)
3. Conduct dimensionality reduction using a feature selection or feature extraction technique such as PCA.
4. Create a model (KNN, Naïve Bayes classifier, or other)
5. Check fitness of the model (e.g. R^2 for regression or % false positive for classification)
6. Update model as new information becomes available.

ASSIGNMENT

Assignment: Which elevator am I in?

Goal. Build a classifier that identifies the elevator from one ride: Hart House, Mechanical Building (MC), Myhal Centre, or Rosebrugh Building.

Part 1: Data collection (groups)

Record at least 20 rides per elevator, so at least 80 rides per group.

Vary the conditions on purpose: up and down, 1 floor and several floors, busy and empty cabins, different times of day.

Log metadata for every ride: building, car, direction, start and end floor, approximate number of people, board orientation, date.

Where a building has several cars, record from more than one, so the classifier learns the building rather than one car.

Write a script that detects each ride's start and stop from the accelerometer and cuts the recording into one segment per ride.

ASSIGNMENT

Part 2: Features
Extract at least 8 features and justify each one physically. For example:

Peak acceleration and peak deceleration. This is the lecture example; note that load affects them.

Jerk: how quickly acceleration ramps up.

Cruise speed, from integrated acceleration or from the barometer's pressure rate (about 0.12 hPa per metre).

Dominant vibration frequencies from an FFT of the signal while cruising.

ASSIGNMENT

Part 3: Classifiers

Implement KNN, naive Bayes, 1D CNN and a decision tree or random forest.

Compare them with a simple heuristic baseline, such as a threshold on peak acceleration.

Apply PCA or wrapper feature selection, and report which features survive.

Part 4: Implementation

Implement the algorithm on the board.

ASSIGNMENT

Deliverable:

1. Create an **upload** algorithm in MATLAB or Python that would classify the occupant status based on the 4 input parameters.

2. Create the firmware for the board and upload the code.

3.Provide a **2-4 page summary** of the method you chose and why.

Include any issues or problems that remain in accurate classification.